

Parts and Wholes

In this chapter we aim at developing an ontological well-founded theory of conceptual part-whole relations.

Parthood is a relation of significant importance in conceptual modeling, being present in practically all conceptual/object-oriented modeling languages (e.g., OML, UML, EER, LINGO). Although it has not yet been adopted as a modeling primitive in the semantic web languages, some authors have already pointed out its relevance for reasoning in description logics (e.g., Lambrix, 2000). Nonetheless, in many of these languages, the concepts of part and whole are understood only intuitively, or are based on the very minimal axiomatization that these notions require. In addition to that, despite of being an active topic in the conceptual modeling literature, there is still much disagreement on what characterizes this relation and about the properties that part-whole relations should have from a conceptual point of view (Saksena et al., 1998; Snoek & Dedene, 2001; Pribbenow, 2002; Opdahl & Henderson-Sellers & Barbier, 2001; Odell, 1998).

Part-whole relations are also fundamental from a cognitive perspective, i.e., for the realization of many important cognitive tasks (Tversky, 1989), and as a foundation for the formalization of other entities that compose our ontology. For these reasons, a theory of parts and wholes is considered as a fundamental part of any *foundational ontology*, regardless if it is a revisionary or a descriptive metaphysics effort (Bunge, 1977; Masolo et al., 2003a; Heller & Herre, 2004).

Theories of parts (Mereologies) have been a central point of interest since the pre-socratic philosophers and along the years many precise theories have been developed. These formal theories provide an important starting point for the understanding and axiomatization of the notion of part, and for this reason they are discussed in section 5.1. Nonetheless, despite of their importance, there are many controversial properties ascribed to the part-whole relation by these theories that cannot be

accepted by cognitive and conceptual theories of parthood. These controversial properties are discussed in section 5.2.

On one hand, formal theories of parts are ontologically extravagant and non-parsimonious, allowing for the existence of entities and for the derivation of transitive relations that are not accepted by cognition. On the other hand, they are too weak to characterize what makes something an integral whole composed of many parts. We thus need a theory of wholes, besides from a theory of parts. For this reason, in section 5.3, we discuss the topic of a *principle of unity* (which relates the parts composing a whole), and its relation to the transitivity of parthood.

Another problematic feature of formal mereologies is the non-differentiation among the roles that parts play within the structure of an aggregate, i.e., perceiving all parts as being of the same type. An important issue in any conceptual theory of parthood is to stipulate the different status that parts can have w.r.t. the whole they compose. For instance: (i) whether objects can share parts; (ii) whether an object *only exists* being part of a specific whole (or of a whole of certain kind); (iii) whether an object *only exists* having a specific object as part (or a part of a specific kind). These different modes of associations, known as *secondary characteristics of parthood*, are discussed in section 5.4.

In section 5.5, we discuss in depth a classical *linguistic* study about meronymic relations, which propose a typology of different types of parthood, as well as a refinement of this study that aims towards a typology of cognitive part-whole relations.

In section 5.6, we demonstrate how the different types of parthood relations elaborated in the typology presented in section 5.5 can be exploited to derive more direct guidelines for prescribing and proscribing transitive parthood relations.

Section 5.7 discusses some relevant related work and section 5.8 elaborates on some final considerations related to the chapter.

5.1 Formal Theories of Parts

The study of parthood relations can be traced back to the early days of philosophy, beginning with the presocratic atomists and continuing throughout the writings of Plato, Aristotle, Leibniz and the early Kant, to cite just a few. The first attempt at a rigorous formulation of the theory was made by Edmond Husserl, (see his third *Logical Investigation* (Husserl, 1970)), but the first completely theory of parts, named *Mereology* (from the Greek μέρος, ‘part’), was proposed in 1916 by the Polish philosopher Stanisław Lesniewski (Lesniewski, 1992), who used the part-whole relation as a substitute for the class membership in standard set theory. This theory

was later elaborated by Leonard and Goodman in their *The Calculus of Individuals* (Leonard & Goodman, 1940).

Lesniewski's *Mereology* and *The Calculus of Individuals* have been used as a common basis for almost all developments in theories of parthood that we have in formal ontology today. An exception is the *Assembly Theory* developed in (Bunge, 1977). We shall also consider Bunge's theory in the analysis that follows, since it has been applied by the conceptual modeling community as a foundation for the part-whole relation in conceptual modeling.

As conveyed by the name *Calculus of Individuals*, these theories describe relations among *individuals*, irrespective of their ontological nature or the meta-level category to which they belong. In other words, the relata can be as different as material bodies, events, geographical regions or abstract entities. Mereologies are formal (i.e., domain independent) theories, which formally characterize the principles underlying the relations between an entity and its constituent parts, just like set theory formally characterizes the underlying relationships between a class and its members.

The presentation and axiomatization of the different mereological theories in this section is based on (Varzi, 1996, 2003), (Simons, 1987) and (Herre & Heller, 2004).

5.1.1 Core Concepts (Minimal Mereology)

In all philosophical theories of parts, including *Lesniewski's mereology*, the *Calculus of Individuals* and the *Assembly theory*, the relation of parthood stands for a partial ordering, i.e., a reflexive, antisymmetric and transitive relation. This is reflected in the following axioms, in which $\leq$ symbolizes the relation of parthood:

- (1). $\forall x (x \leq x)$
- (2). $\forall x, y (x \leq y) \wedge (y \leq x) \rightarrow (x = y)$
- (3). $\forall x, y, z (x \leq y) \wedge (y \leq z) \rightarrow (x \leq z)$

These axioms amount to what is referred in the literature by the name of *Ground Mereology (M)*, which is the core of any theory of parts.

Taking reflexivity (and antisymmetry) as constitutive of the meaning of 'part' implies regarding identity as a limit case of parthood. A stronger *proper part* relation ($<$), whereby nothing counts as part of itself, can thus be defined in terms of this one:

- (4). $(x < y) =_{\text{def}} (x \leq y) \wedge \neg(y \leq x)$

and consequently the following equivalence holds

$$(5) \forall x, y (x \leq y) \leftrightarrow (x < y) \vee (x = y)$$

The *proper part* ($<$) relation is a strict partial ordering, i.e., an asymmetric and transitive relation, from which irreflexivity follows:

$$(6). \forall x \neg(x < x)$$

$$(7). \forall x, y (x < y) \rightarrow \neg(y < x)$$

$$(8). \forall x, y, z (x < y) \wedge (y < z) \rightarrow (x < z)$$

In summary, if we interpret the concept of *being a part* as that of *being a proper part*, we have that for all individuals x, y, z :

- (i) x is not a part of itself;
- (ii) if x is part of y then y is not part of x ;
- (iii) if x is part of y and y is part of z then x is part of z .

Either the *proper part of* or *improper part of* relations can be taken as primitive, and the choice is more a matter of convenience than anything else. While the former is a more natural concept, the latter is algebraically more convenient.

Another important relation in theories of parts is that of *overlapping*. We say that two individuals *overlap* ($\bullet$) if they have a part in common. This also includes the case in which one is part of another and also the case of identity. Overlap is, hence, reflexive and symmetric but not transitive:

$$(9). (x \bullet y) =_{\text{def}} \exists z (z \leq x) \wedge (z \leq y)$$

$$(10). \forall x (x \bullet x)$$

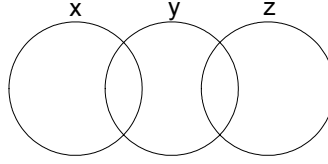
$$(11). \forall x, y (x \bullet y) \rightarrow (y \bullet x)$$

When two individuals overlap but neither is a part of the other then we say that they *properly overlap* ($\circ$):

$$(12). (x \circ y) =_{\text{def}} (x \bullet y) \wedge \neg(x \leq y) \wedge \neg(y \leq x)$$

The example of a proper overlap depicted in figure 5.1 below makes it clear that overlapping is the mereological counterpart of the intersect relation in set theory.

Figure 5-1 Example of a proper overlap between individuals x and y, and y and z



If (and only if) two individuals x and y do not overlap, they are said to be *disjoint*. In other words, x and y are disjoint iff they have no part in common (e.g., x and z in figure 5.1). Disjointness ($\int$) can be defined as follows:

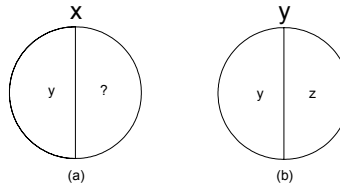
$$(13). (x \int y) =_{\text{def}} \neg(x \bullet y)$$

Like overlapping, disjointness is symmetric

$$(14). \forall x, y (x \int y) \rightarrow (y \int x)$$

The axioms (1-3) define the minimal (partial ordering) constraints that every relation must fulfill to be considered a part of relation. Although necessary, these constraints are not sufficient, i.e., it is not the case any partial ordering qualifies as a parthood relation. For example, the diagram of figure 5.2.a depicts a situation with only two objects x and y , such that y is a proper part of x . This represents a model of the *ground mereology* (in the logical sense), but which can hardly be considered a part of relation, since whenever an object has a proper part, it has more than one, as in the case depicted in figure 5.2.b. The question mark in figure 5.2.a, thus, represents an inexistent proper part of x that would be assumed to exist were y to be considered a part of x .

Figure 5-2 (a) Example of a model that does not satisfy the *weak supplementation axiom* (15); (b) Example of a model of *minimal mereology*



Models such as the one of figure 5.2.a are thus excluded by the following additional constraint:

$$(15). \forall x, y (y < x) \rightarrow \exists z (z < x) \wedge (z \int y)$$

Formula (15) is termed in the literature the *weak supplementation principle*, and the theory composed by (1-3) and (15) is named the *Minimal Mereology* (MM). Some authors (e.g., Simons, 1987), regard (15) as

constitutive of the meaning of part and, hence, consider (1-3) plus (15) as the minimal constraints that a mereological theory should incorporate.

Traditionally, MM has been extended in two different ways, both of which have been subject to philosophical controversy. These two extensions are discussed in the sequel.

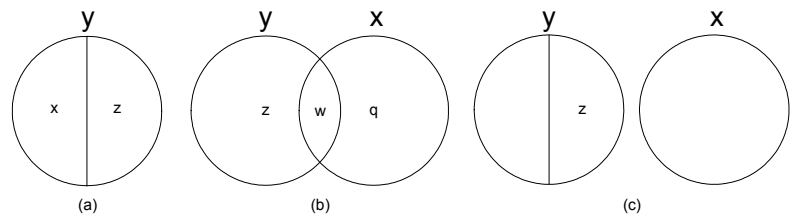
5.1.2 From Minimal to Extensional Mereology

The first extension to MM has been created by strengthening the supplementation principle represented by (15). In this system, (15) is thus replaced by the following stronger supplementation axiom:

$$(16). \forall x, y \neg(y \leq x) \rightarrow \exists z (z \leq y) \wedge (z \int x)$$

This axiom states that if an individual y is not a part of another individual x then there is a part of y which does not overlap with x . This includes the cases depicted in figure 5.3. In both figures 5.3.a ($x < y$) and 5.3.b ($y \circ x$), there is a part z of y , which is disjoint (does not overlap) with x . In figure 5.3.c ($y \int x$), every part of y is disjoint with x .

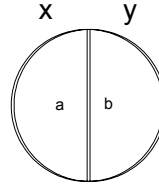
Figure 5-3 Examples of models of *extensional mereology*. In (a), x is a proper part of y ; in (b), x and y properly overlap; in (c), x and y are disjoint. In each of these cases, y is not a part of x and, therefore, there is always a part of y that does not overlap with x .



Formula (16) is named the *strong supplementation principle*, and the theory that incorporates (1-3), (15) and (16) is named *Extensional Mereology (EM)*.

If the axioms of ground mereology are assumed (1-3), then (16) implies (15). However, the converse is not true. For instance, the situation represented in figure 5.4 depicts a MM-model that is not a valid EM-model. In this example, x and y are two distinct entities (symbolized by the double line) but that are composed of the same parts. Every part of x (and y) is supplemented by the other one so that (15) is satisfied. However, the same does not hold for (16), i.e., y is not a part of x , but there is no part of y that is disjoint with x .

Figure 5-4 Two distinct entities that are composed of the same parts; example of a model of *minimal mereology* that is not a model of *extensional mereology*



The name *Extensional Mereology* is exactly motivated by the exclusion of countermodels such as the one figure 5.4. In fact, the following is a theorem of EM:

$$(17) \exists z (z < x) \rightarrow (\forall z ((z < x) \rightarrow (z < y)) \rightarrow (z \leq y))$$

from which it follows that

$$(18) \exists z (z < x) \vee (z < y) \rightarrow ((x=y) \leftrightarrow \forall z ((z < x) \leftrightarrow (z < y)))$$

Theorem (18) states that two objects are identical iff they have the same (proper) parts³⁰. This is the mereological counterpart of the extensionality principle in set theory, which states that two sets are identical iff they have the same members. The philosophical controversy of EM arises exactly because of this axiom. In a *multiplicationist* ontology (Masolo et. al, 2003a) such as the one proposed here, we can have continuants that share the same parts but which are not identical. To use an example already mentioned in chapter 4, a statue and a lump of clay can be composed of the same parts. They are, nonetheless, diverse, since they possess incompatible meta-properties. We return to this point in section 5.2.2.

5.1.3 From Extensional to Classical Mereology

A second way that MM has been extended is with the aim to provide a number of closure operations to the mereological domain. As discussed, for example, in (Varzi, 1996, 2003) and (Simons, 1987), theories named *CMM* (*Closure Minimal Mereology*) and *CEM* (*Closure Extensional Mereology*) can be obtained by extending MM and EM with the following operations:

- a) **Sum:** also named *mereological fusion* (or *juxtaposition* in the case of Assembly Theory). The sum z of two objects x and y , symbolized as $\text{Sum}(z, x, y)$, is the entity such that every object that overlaps with z , overlaps either with x or with y (or with both);

³⁰An analogue for the improper part-of is already provable in M due to the reflexivity and antisymmetry of $\leq$.

$$(19) \text{Sum}(z, x, y) =_{\text{def}} \forall w ((w \bullet z) \leftrightarrow ((w \bullet x) \vee (w \bullet y)))$$

- b) **Product:** also named *superposition* (in Assembly Theory). The product of two objects x and y is the entity z such that every part of z is either part of x or y ;

$$(20) \text{Pro}(z, x, y) =_{\text{def}} \forall w ((w \leq z) \leftrightarrow ((w \leq x) \wedge (w \leq y)))$$

- c) **Difference:** the difference of two objects x and y is the entity z such that every part of z is part of x and does not overlap with y ;

$$(21) \text{Dif}(z, x, y) =_{\text{def}} \forall w ((w \leq z) \leftrightarrow ((w \leq x) \wedge \neg(w \bullet y)))$$

- d) **Complement:** the complement of an entity x is the entity z such that every part of z does not overlap with x ;

$$(22) \text{Comp}(z, x) =_{\text{def}} \forall w ((w \leq z) \leftrightarrow \neg(w \bullet x))$$

These operations are the mereological counterpart of the set theoretical operations of *union*, *intersection*, *set difference* and *complement of a set*, respectively. In the presence of the extensionality principle, the z 's that are the results of these operations are unique. Thus, for example, in an extensional mereology, if two objects x and y overlap then there is a unique entity z that is composed of the common parts of x and y . In particular, the existence of the product and difference of two individuals x and y , and of the complement of an individual x , are only guaranteed in certain cases. These conditions are expressed in formulas (23-25) below, respectively.

$$(23) \forall x, y (x \bullet y) \rightarrow \exists z \forall w ((w \leq z) \leftrightarrow ((w \leq x) \wedge (w \leq y)))$$

$$(24) \forall x \exists y \neg(y \leq x) \rightarrow \exists z \forall w ((w \leq z) \leftrightarrow ((w \leq x) \wedge \neg(w \bullet y)))$$

$$(25) \forall x \exists y \neg(x \bullet y) \rightarrow \exists z \forall w ((w \leq z) \leftrightarrow \neg(w \bullet x))$$

The mereological sum (Sum), conversely, is guaranteed by the presence of an entity, termed the Universe of which everything is part (Simons, 1987):

$$(26) \text{Universe}(z) =_{\text{def}} \forall x (x \leq z)$$

Once more, in an extensional mereology, the universe z is unique. The existence of a “null individual” that is part of everything would also guarantee the existence of the product and difference for any two

individuals, and the existence of a complement for any individual. However, most theories do not define such an entity. An exception is Bunge's Assembly theory (Bunge, 1977).

As demonstrated in (Simons, 1987), if the product operator is functional then the strong supplementation axiom (16) is implied by the weak supplementation principle (15). As a consequence, CMM and CEM collapse in one single theory.

Finally, traditionally, unrestricted operations of fusion and product are also defined for closure mereologies. For the (unrestricted) *mereological sum* we define the following formula schema:

$$(27) \exists x F(x) \rightarrow \exists z \forall y ((y \bullet z) \leftrightarrow \exists w (F(w) \wedge (y \bullet w)))$$

This expresses that for every satisfied predicate F there is an entity consisting of all those things that satisfy F or, to put it differently, z is the sum of the arbitrary non-empty set of entities w_i such that $F(w_i)$ holds. Once more, in the presence of extensionality (16), the entity which is the sum of all entities satisfying predicate F has its uniqueness guaranteed. In this case, we can define a general sum as:

$$(28) \sigma x F(x) =_{\text{def}} \iota z \forall y ((y \bullet z) \leftrightarrow \exists w (F(w) \wedge (y \bullet w)))$$

The product of all members of a set G of overlapping objects can be defined as follows: Let W be set of all those things that are part of every member of set G , i.e.,

$$(29) \forall x W(x) \leftrightarrow \forall y (G(y) \rightarrow (x \leq y))$$

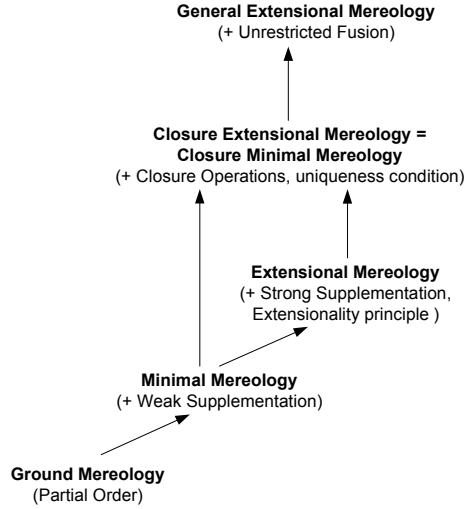
The unrestricted product of all members of set G can, hence, be defined as the sum of all members of W , that is, by replacing F for W in the formula schema (27). The result of adding schema (27) to CMM or CEM is a theory named *GEM* (*General Extensional Mereology*) or *Classical Extensional Mereology*.

As demonstrated, for instance, in (Varzi, 2003), all closure operations (19-22) can be defined via choice of suitable predicates F to be substituted in (27). This gives the full strength of GEM, which has the algebraic structure of a quasi-boolean algebra (boolean algebra with a zero element removed)³¹.

Figure 5.5 below represents schematically the logical space of these different mereological theories.

³¹ In the case of Bunge's Assembly theory, it is as expressive as a complete Boolean lattice.

Figure 5-5 Relations between different mereological theories. The arrows go from the weaker to the stronger theory



A final way to extend the theories in figure 5.5 is by considering the issue of *Atomism*. A *mereological atom* is an entity that has no proper parts:

$$(30) \text{At}(x) =_{\text{def}} \neg \exists y (y < x)$$

In an atomistic mereology everything has atomic parts, i.e.,

$$(31) \forall x \exists y \text{At}(y) \wedge (y \leq x)$$

Conversely, in an atomless mereology the following axiom holds:

$$(32) \neg \exists x \text{At}(x)$$

Formulas (31) and (32) are clearly incompatible, but taken in isolation they can be added to any of the theories depicted in figure 5.5. In other words, the question of atomism is ortogonal and compatible to any of the mereologies discussed so far. Adding (31) to a theory X yields its atomistic version AX, whereas adding (32) yields its corresponding atomless version AX.

Finally, it is important to emphasize two important features of atomistic mereologies:

1. Atomistic mereologies (and only them) admit finite models, i.e., decomposition of parts must eventually come to an end;

2. When atoms are considered, important simplifications can be made to many of the axioms discussed so far (Varzi, 2003). For instance, the *weak supplementation* principle (15) can be replaced by

$$(33) \forall x, y (y < x) \rightarrow \exists z At(z) \wedge (z < x) \wedge \neg(z < y)$$

5.2 Problems with Mereology as a Theory of Conceptual Parts

Mereology has shown itself useful for many purposes in mathematics and philosophy (Varzi, 1996; Simons, 1987). Moreover, it provides a sound formal basis for the analysis and representation of the relations between parts and wholes, and among parts that compose a whole, regardless of their specific nature. However, as pointed out by (Gerlst & Pribbenow, 1995) and (Pribbenow, 2002) (among other authors), it contains many problems that make it hard to directly apply it as a theory of conceptual parts. As it shall become clear in the discussion that follows, on one hand the theory is too strong, postulating constraints that cannot be accepted to hold generally for part-whole relations on the conceptual level. On the other hand, it is too weak to characterize the distinctions that mark the different types of conceptual part-whole relations.

We shall carry out our discussion in a bottom-up approach w.r.t. figure 5.5, i.e., we first discuss the problems of the basic axioms of Ground Mereology, and then continue upwards in the graph discussing the problems of stronger theories.

5.2.1 Conceptual Problems in Ground Mereology

Let us take the proper-part relation, which better reflects our common sense notion of part. According to the axioms (6-8), proper parthood is an irreflexive, antisymmetric and transitive relation. Irreflexivity and antisymmetry are generally accepted as meta-properties that all (proper) part-whole relations should have. However, the same does not hold for transitivity. Take as examples the following cases:

- (i) The hand is part of the arm
The arm is part of the person
Ergo, the hand is part of the person
- (ii) A person is part of the KR (Knowledge Representation) group
The KR group is part of the AI (Artificial Intelligence) group
Ergo, the person is part of the AI group

In these two examples, transitivity holds without a problem. However, putting (i) and (ii) together, the following fallacious argument can be constructed:

- (iii) The hand is part of the person
 The person is part of the KR group
 (?) The hand is part of the KR group

Another example where transitivity fails is the following:

- (iv) Eschede is part of The Netherlands
 The Netherlands is part of the European Union
 (?) Enschede is part of the European Union

In (iii) and (iv), the problem arises because the mereological parthood relation fails in taking into account the different roles that parts play within the whole. In order to do this, we must complement mereology (the theory of parts) with a *theory of wholes* (Gangemi et al., 2001), in which the relations that tie the parts of a whole together are also considered. In section 5.3, we use a theory of integral wholes (Simons, 1987) to address the problem of transitivity in conceptual part-whole relations. We advocate that conceptual parthood relations should not be interpreted in an unconstrained manner, but always in regards to a specific *context*, and that these contexts demarcate the scopes in which transitivity can be guaranteed to hold. Moreover, we claim that understanding and defining what a context is amounts to the same task as understanding what makes something an integral whole. Finally, as we demonstrate in section 5.5, there is not just one, but many distinct notions of conceptual parts, and the issue of transitivity must be considered differently for each of these notions.

Finally, something that becomes clear in section 5.3 is that, although we deal with atomistic theories, the notion of atom as defined in the classical theories is too coarse. When defining contextual parthood, besides from absolute atoms, we need the notion of atoms relative to a given context or level.

5.2.2 Conceptual Problems in Extensional Mereology

The problem with extensional mereologies from a conceptual point of view arises from the introduction of the strong supplementation principle and, consequently, of formula (16) which states that objects are completely defined by their parts. In EM (and all its extensions), two objects are identical iff they have the same (proper) parts. This is the mereological counterpart of the extensionality principle in set theory, and the problems

associated with extensionality in part-whole relations from a conceptual point of view are similar to those associated with equating universals with sets in classical nominalistic theories of universals (Armstrong, 1989).

Take, for example, an organization A composed by the members John, Peter and Mary. Now suppose a different organization B, composed of exactly the same members. According to (16) the following can be observed:

- (i). A and B are identical;
- (ii). If John leaves organization A then the organization A' whose members are now only Peter and Mary is not the same organization as A;
- (iii) since John did not leave organization B, organizations A and B are identical in world w but not in world w'.

In parity with authors such as (Gerlst & Pribbenow, 1995), we believe that these conclusions are unacceptable. First, we can easily imagine two different organizations which share the same members. Imagine the situation in which John Lennon, Paul McCartney, George Harrison and Ringo Star are also the members of the indoor football team Liverpool F.C. The Beatles and the Liverpool F.C. are clearly different individuals despite of sharing the same parts. Not only they might obey completely different principles of identity but also their members play completely different roles within the internal structure of the composite. Using the same example, (ii) is clearly false for the case of The Beatles, since Ringo Star replaced Pete Best without spawning the creation of a new entity (and destruction of the former). Finally, (iii) is simply unacceptable because it contradicts Leibniz's Law (see chapter 4), which is accepted as an axiom in practically all theories of identity (van Leeuwen, 1991).

Examples such as this one can be found for many non-identical entities that happen to coincide in a certain world (e.g., the statue and lump of clay example previously discussed). In summary, extensional theories differentiate entities that in common sense we deem as the same and equate entities that we deem different. Moreover, it rejects the (strongly supported) claim defended in chapter 4 that there are several distinct principles of identity that are employed by human cognition. To accept extensionality amounts to the same as accepting the existence of one single principle of identity, namely, the *extensionality principle*. As discussed in section 4.3, the only period in which humans employ a unique principle of individuation and identity is in the pre-language period when they are younger than 9-months old. In this period, infants' judgment of identity statements is supported by the unique sortal "*maximally-self-connected-physical-object*", i.e., the notion of an object with clear boundary contrasts

with its background and whose parts move along together with the whole (van Leeuwen, 1991, part 3). Perhaps, extensionality holds for this primitive notion of physical object as well as for *amounts of matter* (see section 5.5.1 of this chapter) but it certainly does not hold in general for substantials. Furthermore, as demonstrated by (Xu, 2004), this initial object-based individual system is overridden later in human cognition by a kind-based system with a multitude of sortal-supplied principles of identity.

A different perspective on this problem is that extensionality considers all parts of an entity as essential, i.e., an entity is equal to the mereological sum of its parts, thus, changing any of its parts changes the identity of that entity. Ergo, an entity cannot exist without each of its parts, which is the same as saying that all its parts are essential. This conclusion is clearly false for many objects accounted by common sense. For example, as a person, there are many parts that I can loose without ceasing to be the same. For instance, during a person's lifetime one typically looses hair, teeth and nails. There are also more tragic cases when one can loose a member such as a finger, an arm or a leg. In fact, during a lifetime a person changes all cells in her body! Nonetheless, none of these changes alter the identity of the individual in question. This is not to say that there are no essential parts in the human body. For instance, one cannot survive without a heart or a brain. Additionally, despite that both relations person/heart and person/brain constitute ontological dependence relations, they are of quite diverse nature. Whilst in the former, we have a case of generic dependency; the latter exemplifies a case of specific existential dependency that characterizes real *essential parts*.

In summary, while some parts of a common sense object are essential, not all of them are essential. This topic is thoughtfully discussed in section 5.4.2.

5.2.3 Conceptual Problems in General Extensional Mereology

From a conceptual point of view, the problem with the theory of General (Classical) Extensional Mereology is related to formula schema (27), which defines the existence of a sum (or fusion) for any arbitrary non-empty (but non-necessarily finite) set of entities. Just as in set theory one can create a set containing arbitrary members, in GEM one can create a new object by summing up individuals that can even belong to different ontological meta-categories. For example, in GEM, the individual Θ created by the sum of Noam Chomsky's left foot, the first act of Puccini's Turandot and the number 3, is an entity considered as legitimate as any other.

In the literature, this feature of GEM is termed *ontological extravagance*, referring to the commitment of the theory to the existence of a wealth of entities that are utterly counterintuitive and that have no place in human

cognition. In addition, GEM is also deemed to be *ontologically exuberant*, since it dramatically increases the number of entities to be included in the inventory of the domain (Varzi, 2003).

As argued by (Pribbenow, 2002), humans only accept the summation of entities if the resulting mereological sum plays some role in their conceptual schemes. To use an example cited by Pribbenow: the sum of a frame, a piece of electrical equipment and a bulb constitutes an integral whole that is considered meaningful to our conceptual classification system. For this reason, this sum deserves a specific concept in cognition and name in human language. The same does not hold for the sum of bulb and the lamp's base. Another example of meaningful sums occurs when entities are summed up to provide a target for a linguistic plural reference such as in "two of the children of my sister".

A way to understand why only meaningful sums are accepted by cognition is to recall the purpose of classification in our conceptual system. As discussed in chapter 4, the ability to categorize is central for our construal of the environment, and without it, learning would be hardly possible. For example, if I am able to recognize that *Felix* is an instance of the category Cat, then I am able to infer that *it* has all the properties that I know that apply to cats. Additionally, all properties that I come to learn about *Felix* can be carried out to other cats that I may encounter. Finally, as point out by (Milikan, 1998; Mcnamara, 1986; Schuun & Vera, 1995), our cognitive system seems to be specially tuned to form conceptual categories that are based on a common *essence*, supported by a theory-based explanation for their principle of application (see section 5.8). This is because these are the categories that provide the most useful sources for inductive knowledge. In this perspective, what is the point for our cognitive system to recognize the existence of entities as unique and alien as the mereological sum Θ aforementioned? What properties can we expect Θ to have? What can we learn about it that can be carried to other individuals?

For this reason, we advocate that a theory of conceptual part-whole relations should forbid the construction of unrestricted sums, and formula schema (27) should only be defined for predicates that represent genuine universals (see discussion on chapter 6). In contrast, the notions of *integrity* and *unifying condition* that comprise a theory of wholes should be taken seriously. This is to say that a theory of conceptual parts and wholes should only countenance the existence of composite objects that are unified by bona fide unifying conditions. These issues are discussed in depth in section 5.3 below.

5.3 Integral Wholes

As we have previously discussed, one of the major conceptual problems with Classical Extensional Mereology comes from the generalized fusion axiom, which allows for the existence of a sum (or fusion) for any arbitrary non-empty set of entities. For example, it allows for the definition of an aggregate composed by the state of California and the number 3. A question that comes to the mind is: If these entities are unobjectionable from a formal point of view, why are they deemed unnatural by common sense? Or, equivalently, what makes an aggregation such as car, a football team, a forest, a human body, to be accounted as a *natural conceptual whole*?

According to (Simons, 1987), the difference between purely formal ontological sums and, what he terms, *integral wholes* is an ontological one, which can be understood by comparing their existence conditions. For sums, these conditions are minimal: the sum exist just when the constituent parts exist. By contrast, for an integral whole (composed of the same parts of the corresponding sum) to exist, a further *unifying condition* among the constituent parts must be fulfilled.

The distinction between a mere sum and an integral whole also appears in (Bunge, 1977, 1979) in the form of the distinction between, what he terms, a *mere aggregate* and a *system*. Bunge defines a mere aggregate as “a compound thing, the components of which are not coupled, link, connected or bounded [, and which...] therefore lacks integrity or unity” (Bunge, 1979, p.4). In contrast, the components of system are “interrelated rather than loose”.

Let us start by defining the Bungean concept of a *link*, *coupling* or *bond* among two things. Bunge explicitly states that “[w]e must distinguish between a mere relation, such as that of being older, and a connection, such as that of exerting pressure.... [T]wo things are *connected* (or *coupled*, or *linked*, or *bounded*) if at least one of them acts upon the other...where the action need not consist in eventuating something but ... one thing acts upon the another if it modifies the latter history” (p.6).

Bunge emphasizes the distinction between *the composition of a system* and the merely formal notion of composition: “A social system is a set of socially linked animals. The brains of such individuals are parts of the latter but do not qualify as members of components of a social system because they do not enter independently in social relations: only entire animals can hold social relations. In other words, the composition of a social system is not the collection of its parts but just the set of its *atoms*, i.e., those parts that are *socially connectable*” (ibid., p.5).

The relative composition of a system can thus be defined as follows:

Definition 5.1 (relative composition): Let S be the set of substantial individuals, and let the composition of a system $x \in S$ be defined as

$$(34). C(x) = \{y \mid y \leq x\}.$$

Then the A -composition of a system x is defined as

$$(35). C_A(x) = \{y \in A \mid y \leq x\},$$

or simply, $C_A(x) = C(x) \cap A$. The set $A \subseteq S$ is said to contain the A -atoms of x . ■

Although this notion conceptualizes an important intuition, it begs some important clarifications. To start of with, Bunge is not explicit on how to define the set of atoms A . However, the text makes clear that the elements of A must be those elements in the world that participate in certain *bonding* relations. He uses examples such as: the molecular composition of a mass of water is the set of H_2O molecules that are part of that mass, which, hence, excludes individual Hydrogen and Oxygen atoms. Alternatively, one could say that the molecular composition of a mass of water is composed by the set of parts of that mass that can engage in molecular relations. An analogous argument can be made in the case of a social system aforementioned: the set of social atoms of a system is composed by those parts of the system that can engage in social relations. For instance, the social composition of a school is the set of staff and pupils, which are held together by relations of teaching and learning, managing and being managed, among others (Bunge, *ibid.*, p.5).

Some important notions that can be used to provide a foundation for the set of A -atoms of a system have been proposed by the philosopher Peter Simons in (Simons, 1987), namely, the notions of a *R-closed*, *R-connected* and *R-closure system*.

Definition 5.2 (R-closed): A set B is closed under a relationship R , or simply R -closed, iff

$$(36). cl \langle R \rangle B =_{\text{def}} \forall x (x \in B) \rightarrow ((\forall y R(x,y) \vee R(y,x) \rightarrow (y \in B)),$$

i.e., the relationship R does not cross B 's borders in any direction. ■

Definition 5.3 (R-connected): A set B is connected under the relation R , or simply, R -connected iff

$$(37). \text{con } \langle R \rangle B =_{\text{def}} \forall x (x \in B) \rightarrow (\forall y (y \in B) \rightarrow (R(x,y) \vee R(y,x))),$$

i.e., every member of B bears a relation with another member of B . ■

Definition 5.4 (R-closure system): A set B is a closure system under the relation R , or simply, R -closure system iff

$$(38). \text{cs } \langle R \rangle B =_{\text{def}} (\text{cl } \langle R \rangle B) \wedge (\text{con } \langle R \rangle B),$$

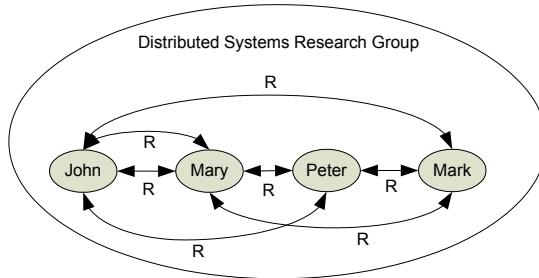
i.e., iff it is both connected and closed under R . ■

Definition 5.5 (A-parthood): Let the set of A -atoms of a system x be defined in terms of a relationship R , such that A is a set of parts of x that form a closure system under R . In this case, x is said to be an *integral whole unified under relationship R* , and R is termed a *characteristic relation* or *unifying condition* for x . Therefore, we can finally define a parthood relation relative to A , or simply, an A -part as

$$(39). (y \leq_A x) =_{\text{def}} (y \in C_A(x)).$$

An example of an integral whole is depicted in figure 5.6 below. The unifying relation R , in this case, is the relation of *carrying out research in the area same area in the University of Twente*. In this figure, the parts of the Distributed Systems Research Group (DSRG) that are not R -connected are excluded from its relativized composition (e.g., all the anatomical parts of John, the cellular parts of these anatomic parts, the atomic parts of these cellular parts, etc.).

Figure 5-6 Example of an *integral whole* unified by a characteristic relation R

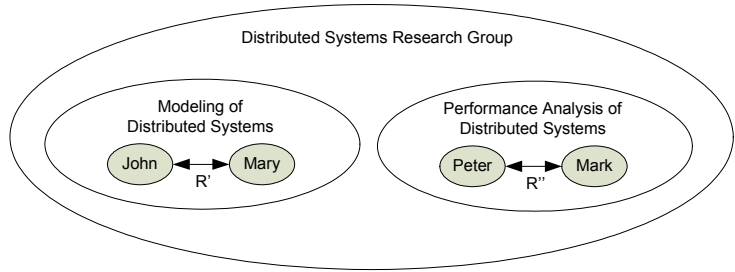


Bunge also provides no explanation for why the members of A are to be considered atoms. However, according to the interpretation just given for

the origin of set A , its members must indeed be A -atomic, i.e., if $\forall x (x \in A) \rightarrow \neg \exists y (y <_A x)$ ³². This is because, the whole x unified under R is maximal under this relation, by the definition of an R -closure system. As a consequence, no member of A can be unified under the same relation R .

This is far from saying that they must be atomic w.r.t. the formal parthood relation. In fact, there can be other relations R' and R'' , such that they can be used to form other closure systems among the subsets of A . Suppose that R' is the relation of *carrying out research in the same sub-area of the area of distributed systems in the University of Twente*, we can then form the set B which is an R' -closure system. Since every member of B is a formal part of the Distributed Systems Research Group (DSRG), we can say that the individual *Modeling of Distributed Systems Research Group* (MDSRG) is an integral whole unified under R' , and also a part of DSRG. An analogous procedure can be made for relation R'' creating the individual *Performance Analysis of Distributed Systems Research Group* (PADSRG). This situation is depicted in figure 5.7.

Figure 5-7 Refinements of a unifying relation R providing further structure within a whole



The set of B -atoms under R' is $\{\text{John}, \text{Mary}\}$, and the set of C -atoms under R'' is $\{\text{Peter}, \text{Mark}\}$. Now, let us redefine the set of A -atoms under R as $\{\text{John}, \text{Mary}, \text{Peter}, \text{Mark}, \text{MDSRG}, \text{PADSRG}\}$. It is clear that DSRG can be unified under R , since: (i) every member of A is part of DSRG; (ii) every member of A is connected under R (e.g., John carries research in the same area as the MDSRG, which in turn, carries research in the same area as the PADSRG). The relations R' and R'' are specializations of R and, thus, the sets B , C and A defined as closure systems of R' , R'' and R , respectively, are such that $B \subseteq A$ and $C \subseteq A$.

Now, it is clear that $\neg(\text{John} <_A \text{MDSRG})$, since MDSRG is an A -atom. However, it is true that $(\text{John} <_B \text{MDSRG})$, $(\text{MDSRG} <_A \text{DSRG})$ and $(\text{John} <_A \text{DSRG})$. In fact, it is generally the case that if a characteristic relationship R' is an specialization of another characteristic relationship R ,

³² The relative proper parthood can be defined as usual, i.e., $(y <_A x) =_{\text{df}} (y \leq_A x) \wedge \neg(y = x)$.

and if the sets B and A are unified under R' and R, respectively, then it is true that

$$(40). \forall x,y,z (x <_B y) \wedge (y <_A z) \rightarrow (x <_A z).$$

Proof: Suppose that $\neg(x <_A z)$. Since, $(x <_B y)$ we have that $(x < y)$ and $(x \in B)$, by the definition of $<_B$. Likewise, since $(y <_A z)$ then $(y < z)$ and $(y \in A)$. From $(x < y)$ and $(y < z)$ we have $(x < z)$, since formal parthood is always transitive. Moreover, since R' is a unifying subrelationship of R then $B \subseteq A$. Therefore, $(x \in A)$, and, consequently, $(x \in C_A(z))$, by definition 5.1. Finally, we have that $(x <_A z)$, which contradicts the hypothesis. □

We can now define the concept of a context as follows:

Definition 5.6 (Context): Let $(R_i \prec R_j)$ mean that if I is a R_i -closure system and J is an R_j -closure system then $(I \subseteq J)$. Let $(R_1 \prec \dots \prec R_n)$ be a series of unifying relations, and let $(\leq_1, \dots, \leq_n)$ be a series of relative part-whole relations defined such as in definition 5.5. We name $(\leq_1, \dots, \leq_n)$ a context, and, by a generalization of (40), we have that

$$(41). \forall x,y,z (x \leq_i y) \wedge (y \leq_{i+1} z) \rightarrow (x \leq_{i+1} z). \quad \blacksquare$$

According to definition 5.6, transitivity holds generally within a context. However, for some types of part-whole relations there are no possible unifying subrelationships and, therefore, it is senseless to discuss contextual transitivity. For instance, the relationship *being-a-direct-functional-part*, by definition, cannot have a unifying subrelationship. Otherwise, it would have an “indirect direct-functional-part”.

The discussion on the problem of transitivity carried out in this section is just a preliminary account of the relation between unifying relations for integral wholes and the transitivity of their parts. In sections 5.6 and 7.4, we are able to elaborate on this discussion by considering other distinctions of our ontological theory.

5.4 Secondary Properties of Part-Whole relations

In this section, we focus on some important aspects of conceptual part-whole relations, which are not taken into account in classical mereological theories. Besides the classical notion of parthood defined, for example, by the axioms of Minimal Mereology, there are other axiom groups that can be

used to formally characterize further ontological distinctions among part-whole relations. Two of these so-called *secondary characteristics* (Opdahl & Henderson-Sellers & Barbier, 2001) are discussed as the following subsections. In subsection 5.4.1, we discuss *shareability*, whereas *separability* is discussed in 5.4.2.

5.4.1 Shareable Parts

Shareability is considered a secondary characteristic of part-whole relations, in the sense that it is not the case that it is a property held by all relations of this type. In contrast, it can be used as *differentiae* among distinct types of parthood. Shareability is simpler to understand than the other secondary characteristics discussed in section 5.4.2, since it does not involve modal properties. A first tentative definition of a non-shareable (exclusive) part is given as follows:

Definition 5.7 (exclusive part – first version): An individual x is said to be an exclusive (proper) part of another individual y (symbolized as $x <_x y$) iff (i) x is a (proper) part of y ; (ii) for every z such that x is a (proper) part of z then either: (a) z is a (improper) part of y or (b) y is a (improper) part of z , or (c) both, i.e., y and z are identical.

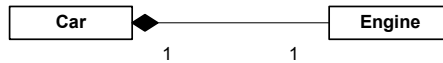
$$(42). (x <_x y) =_{\text{def}} (x < y) \wedge (\forall z (x < z) \rightarrow (z \leq y) \vee (y \leq z))$$

■

From x being an exclusive part of y it does not follow that x is not part of anything else, as it is sometimes wrongly stated in the literature of conceptual/object-oriented modeling. This is because, due to transitivity, x must also be part of everything y is part of.

An example of an exclusive parthood is represented in figure 5.8. In this picture, every instance of an engine is an exclusive part of a car, i.e., if the engine is also part of some other object (e.g., the power system of the car, not represented) then this object must also be part of the car, or it must have the car as one of its parts.

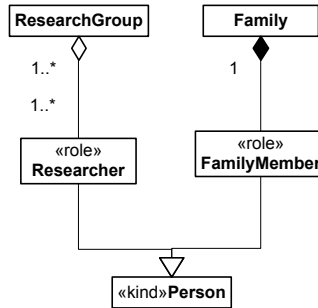
Figure 5-8 An exclusive (i.e., non-shareable) parthood relation represented by the UML composition meta-construct



At first, definition 5.7 seems to suitably represent what it is meant by exclusiveness of parthood in conceptual modeling. However, on second thought, it becomes clear that it is too restrictive to figure as the only definition of exclusiveness to be employed. Take, for instance, the model of figure 5.9. This is an example that represents a recurrent pattern in

conceptual models. Since the identity of the roles researcher and family member are supplied by the substance sortal person that they both subsume, it can certainly be the case that the same individual, say the person John Smith, is both a researcher and a family member and, therefore, is part of both a family and a research group. The logical model corresponding to this specification would be excluded by definition 5.7.

Figure 5-9 Conflict in the notion of exclusiveness. Examples of exclusive and shareable parthood relations represented by the UML meta-constructs of composition (black diamond) and aggregation (white diamond), respectively.



In order to remedy this situation, we propose a much weaker type of parthood exclusiveness, more adequate for the purposes of conceptual modeling. In the definition that follows, we use the notation $x::U$ to represent the relation of instantiation between an individual x and a universal U .

Definition 5.8 (exclusive part): An individual x of type A is said to be an exclusive (proper) part of another individual y of type B (symbolized as $<_x(x,A,y,B)$) iff y is the only B that has x as part.

$$(43). <_x(x,A,y,B) =_{\text{def}} (x::A) \wedge (y::B) \wedge (x < y) \wedge (\forall z (z::B) (x < z) \rightarrow (y = z))$$

■

This type of exclusive part-whole relation can be also defined between the universals A and B as a general exclusive part-whole relation.

Definition 5.9 (general exclusive part-whole relation): A universal A is related to a universal B by a relation of general exclusive parthood (symbolized as $A <_{\text{GX}} B$) iff every instance x of A has an exclusive part of type B .

$$(44). A <_{\text{GX}} B =_{\text{def}} \forall x (x::A) \rightarrow \exists y (y::B) \wedge <_x(x,A,y,B)$$

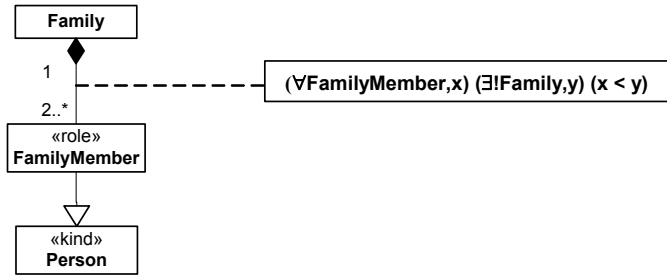
or simply,

$$(45). A <_{GX} B =_{\text{def}} \forall x (x::A) \rightarrow \exists! y (y::B) \wedge (x < y)$$

■

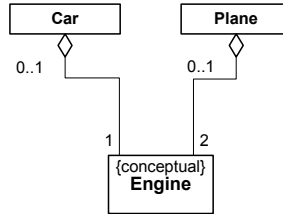
Figure 5.10 shows part of the example of figure 5.9 by using definition 5.9. In the specification of figure 5.10, we employ a language with sortal restriction such as the languages L_1 and L_2 defined in section 4.4.

Figure 5-10
Exclusiveness defined in
the type level and
corresponding
axiomatization



In the articles (Saksena & France & Larrondo-Petrie, 1998) and (Saksena et al., 1998), Monika Saksena and colleagues have proposed a different notion of shareability, which they term *conceptual shareability*. This idea is exemplified in figure 5.11, which is an exact copy from the original model in (Saksena et al., 1998)³³.

Figure 5-11 Conceptual
shareable parthood
relation (from Saksena
et al., 1998)

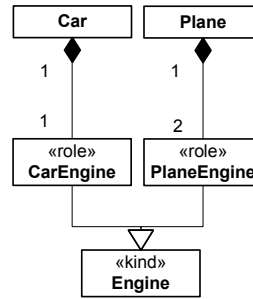


The intention behind the model of figure 5.11 is to represent that although an instance of an engine cannot be part of both a car and a plane, “the concept represented by the class is shared” (Saksena et al., 1998, p.4). Here we defend the idea that this alleged type of shareability is not only superfluous, but also stimulates the creation of poorer conceptual models. As we have discussed in section 4.5.3, conceptual models without minimum cardinality constraints equal to zero can be considered more suitable both in terms of ontological adequacy, and of practical performance in problem-solving tasks. In a situation such as the one

³³ The tagged value {conceptual} in the picture is a proposal of (Saksena et al., 1998).

depicted in figure 5.11, if an engine is built to be part of a car then it is a car engine, if it is built to be part of a plane, then it is a plane engine. Additionally, if the same engine type can be used both in cars and planes then, for every individual engine, being part of a car (or plane) is merely a contingent fact. In other words, every engine that is part of a car in world w could become (or have been) part of a plane in a world w' (or vice-versa). Therefore, in the latter case, both concepts car engine and plane engine must be conceived as roles that an engine can play in certain circumstances. In figure 5.12, we present a version of this latter interpretation of figure 5.11, in which these roles, which are implicit in figure 5.11, are made explicit.

Figure 5-12 The alleged *Conceptual Sharability* relation modelled by real non-shareable relations and explicitly represented roles



5.4.2 Essential and Mandatory Parts

As discussed by (Simons, 1987), there are many issues regarding part-whole relations that cannot be clarified without considering *modality*. One of these issues is the secondary characteristic of *separability*.

In order to formally define separability, we first define some notions related to the topic of *ontological dependence*. In particular, the relations of existential and generic dependence discussed in the sequel are strongly based on those defined in (Husserl, 1970).

Definition 5.10 (existential dependence): Let the predicate $\mathcal{E}$ denote existence. We have that an individual x is *existentially dependent* on another individual y (symbolized as $ed(x,y)$) iff, as a matter of necessity, y must exist whenever x exists, or formally

$$(46). ed(x,y) =_{\text{def}} \Box(\mathcal{E}(x) \rightarrow \mathcal{E}(y))$$

■

From a philosophical standpoint, there is a problem with this definition, namely, that it makes all objects existentially dependent of necessarily existing things (e.g., numbers or platonic forms). Therefore, a better definition would be $ed(x,y) =_{\text{def}} \neg \Box \mathcal{E}(y) \wedge \Box (\mathcal{E}(x) \rightarrow \mathcal{E}(y))$. Nonetheless, for the common-sense contingent individuals that we have interest in conceptual modeling, this difficulty can be overseen, since we are mostly interested here in parthood relations between substantial individuals. For this reason, and for the sake of clarity in the presentation, the safeguard condition that rules out necessary existence ($\neg \Box \mathcal{E}(y)$) is omitted in the subsequent definitions.

With definition 5.10 we can define the concept of an essential part as follows:

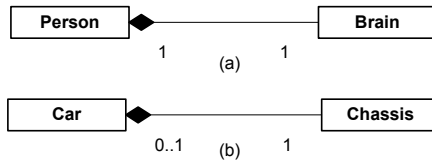
Definition 5.11 (essential part): An individual x is an essential part of another individual y iff, y is existentially dependent on x and x is, necessarily, a part of y : $EP(x,y) =_{\text{def}} ed(y,x) \wedge \Box (x \leq y)$. This is equivalent to stating that $EP(x,y) =_{\text{def}} \Box (\mathcal{E}(y) \rightarrow \mathcal{E}(x)) \wedge \Box (x \leq y)$, which is, in turn, equivalent to $EP(x,y) =_{\text{def}} \Box (\mathcal{E}(y) \rightarrow \mathcal{E}(x) \wedge (x \leq y))$. We adopt here the *mereological continuism* defended by (Simons, 1987), which states that the part-whole relation should only be considered to hold among existents, i.e., $\forall x,y (x \leq y) \rightarrow \mathcal{E}(x) \wedge \mathcal{E}(y)$. As a consequence, we can have this definition in its final simplification

$$(47). EP(x,y) =_{\text{def}} \Box (\mathcal{E}(y) \rightarrow (x \leq y))$$

■

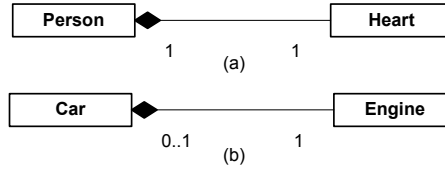
Figures 5.13.a and 5.13.b below depict examples of essential parts. In figure 5.13.a, every person has a brain as part, and in every world that the person exists, the very same brain exists and is a part of that person. In figure 5.13.b, we have an analogous example: a car has a chassis as an essential part, thus, the part-whole relation between car and chassis holds in every world that the car exists. To put in a different way, if the chassis is removed, the car ceases to exist as such, i.e., it loses its identity.

Figure 5-13 Wholes and their Essential parts



The UML notation used in figure 5.13 highlights a problem that exists in practically all conceptual modeling languages. In order to discuss this problem, let us examine another model represented in figure 5.14.

Figure 5-14 Wholes and their Mandatory parts



According to the UML semantics, the models of figure 5.13.a and 5.14.a convey exactly the same kind of information. However, this is not the case, in general, in this domain in reality. Typically, the relation between a person and his brain is not of the same nature as the relation between a person and his heart. Differently from the former, a particular heart is not an essential part of a person, i.e., it is not the case that for every person x there is a heart y , such that in every possible circumstance y is part of x . For instance, the fact that an individual John had the same heart during his entire lifetime was only accidental. With the advent of heart transplants, one can easily imagine a counterfactual in which John had been transplanted a different heart. An analogous argument can be made in the case of figure 5.13.b. Although every car needs an engine, it certainly does not have to be the same engine in every possible world.

The difference in the underlying real-world semantics in the cases of figure 5.13.a and 5.14.a are made explicit if we consider their corresponding formal characterization. In the case of fig.5.13.a, since it is a case of essential parthood, we have that:

(figure 5.13.a) $\Box ((\forall \text{Person}, x)(\exists ! \text{Brain}, y) \Box (\mathcal{E}(x) \rightarrow (y < x)))$

Whereas in the case of figure 5.14.a, the corresponding axiomatization is

(figure 5.14.a) $\Box ((\forall \text{Person}, x) \Box (\mathcal{E}(x) \rightarrow (\exists ! \text{Heart}, y)(y < x)))$

A similar distinction can be made for the case of figures 5.13.b and 5.14.b:

(figure 5.13.b) $\Box ((\forall \text{Car}, x)(\exists ! \text{Chassis}, y) \Box (\mathcal{E}(x) \rightarrow (y < x)))$

(figure 5.14.b) $\Box ((\forall \text{Car}, x) \Box (\mathcal{E}(x) \rightarrow (\exists ! \text{Engine}, y)(y < x)))$

In cases such as those depicted in the specifications of figures 5.13.b and 5.14.b, an individual is not specifically dependent of another individual, but *generically dependent* of any individual that instantiates a given universal. The concept of generic dependence is defined as follows:

Definition 5.12 (generic dependence): An individual y is *generic dependent* of a universal U iff, whenever y exists it is necessary that an instance of U exists. This can be formally characterized by the following formula schema:

$$(48). \text{GD}(y, U) =_{\text{def}} \Box(\mathcal{E}(y) \rightarrow \exists U, x \mathcal{E}(x))$$

■

We name individuals such as the instances of Heart and Engine in figures 5.13.b and 5.14.b, respectively, *mandatory parts*.

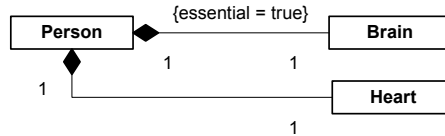
Definition 5.13 (mandatory part): An individual x is a *mandatory part* of another individual y iff, y is generically dependent of an universal U that x instantiates, and y has, necessarily, as a part an instance of U :

$$(49). \text{MP}(U, y) =_{\text{def}} \Box(\mathcal{E}(y) \rightarrow (\exists U, x)(x < y)).$$

In order to represent the ontological distinction between essential and mandatory parts, we propose an extension to the UML notation used in the examples for the remaining of this chapter. We assume that the minimum cardinality of **1** in the association end corresponding to the part represents a *mandatory part-whole* relation. To represent the case of an *essential part-whole* relation, we propose to extend the current UML aggregation notation by defining the Boolean meta-attribute **essential**.

When the meta-attribute *essential* equals *true* then the minimum cardinality in the association end corresponding to the part must also be **1**. This is expected to be the case, since essential parthood can be seen as a limit case of mandatory parthood. When *essential* equals *false*, the tagged value textual representation can be omitted. This extended notation is exemplified in figure 5.15 below.

Figure 5-15 Extensions to the UML notation to distinguish between *essential* and *mandatory* parts



We emphasize that the particular examples chosen to illustrate the distinction between *essential* and *mandatory* parts are used here for illustration purposes only. For example, when modeling *brain* as an essential part of persons and *heart* as a mandatory one, we are not advocating that this is a general ontological choice that should be countenanced in all conceptualizations. Conversely, the intention is to make explicit the

consequences of this modeling choice, and to advocate for the need of explicitly differentiating between these two modes of parthood. The choice itself, however, is always left to the model designer and is conceptualization-dependent. For example, an ontological choice implicit in the assumption of brain/heart example is that brain transplants are not possible and that the identity of an individual person is determined by the identity of her brain. A discussion on the issue of *personal identity* falls outside the scope of this work, and we are not in position here to discuss the possibility of brain transplants. Again, these ontological choices are taken here merely for the purpose of exemplification. The important point is that there are essential and mandatory parts in the world and one should be able to explicitly and suitably model them.

By using the definition of essential parts, one can formally characterize the notion of those individuals prescribed by Extensional Mereologies (EM), i.e., those individual for which all their parts are essential. These entities, named here *extensional individuals* are formally defined in the sequel.

Definition 5.14 (extensional individual): An individual y is named an extensional individual iff for every x such that x is a part of y , x is an essential part of y :

$$(50). E(y) =_{\text{def}} \Box (\forall x (x < y) \rightarrow EP(x,y))$$

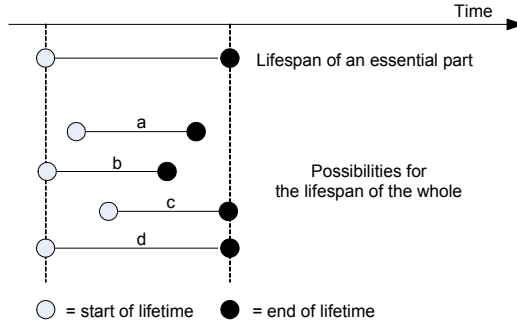
■

Our criticism to EM in section 5.2.2 was that this theory treats all individuals as extensional. This is not to say that there are no such entities. Examples of extensional individuals are discussed in section 5.5.

Up to this moment, we have interpreted possible worlds as maximal state of affairs, which can be factual or counterfactual. In other words, we have assumed a branching structure of time, and each world is taken as a time interval in a (factual or counterfactual) time branch. An alternative is to interpret possible worlds as histories, i.e., as the sum of all state of affairs in a given time branch. In this alternative conception of worlds, we can examine the possible relations between the lifespan of wholes and parts in different types of parthood relations. For instance, figure 5.16 illustrates the possible relations between the lifespan of a whole and one of its essential parts³⁴.

³⁴Actually, figure 5.16 depicts the possibilities for the relation between the lifespan of an object x (top) and any object y which is existentially dependent on x .

Figure 5-16 Possible relations between the life spans of an individual whole and one of its *essential parts*



This figure illustrates the true possibilities for, for instance, the relation between a chassis and a car as depicted in figure 5.13.b. In this case, the lifetime of the chassis is completely independent from the lifetime of any of the cars it happens to be a part of. Actually, as represented in figure 5.13.b, a chassis does not even have to be connected to a car (whole). This is a case of, what we term, *essential part with optional whole*.

Conversely, if we analyze the relation between a brain and a person, we come to the conclusion that the lifespan (d) in figure 5.16 is the only real possibility in this case. That is to say that the lifespan of a person and her brain should necessarily coincide. This is because, in this case, a brain is also existentially dependent on its host. Whenever we have the situation that a part is existentially dependent on the whole it composes, we name it an *inseparable part*.

Definition 5.15 (inseparable part): An individual x is an inseparable part of another individual y iff, x is existentially dependent on y , and x is, necessarily, a part of y :

$$(51). \text{IP}(x,y) =_{\text{def}} \Box(\mathcal{E}(x) \rightarrow (x \leq y))$$

■

The possible relations between the life spans of an inseparable part and its (essential) whole are depicted in figure 5.17. The case of an essential and inseparable part is shown in figure 5.18.

Figure 5-17 Possible relations between the life spans of an individual whole and one of its *inseparable parts*

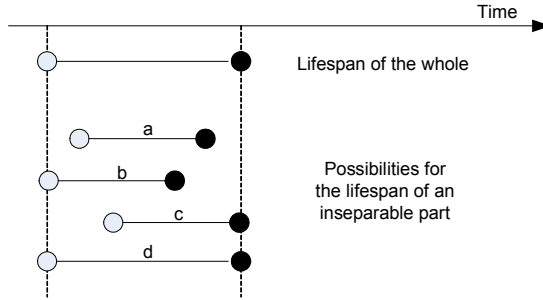


Figure 5-18 Possible relations between the life spans of an individual whole and one of its *essential and inseparable parts*

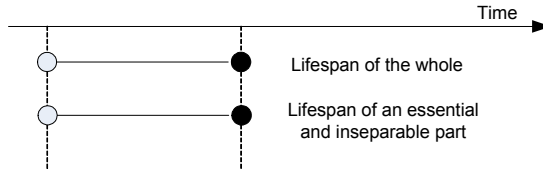


Figure 5.17 does not represent all the possibilities for, for instance, the relation between a heart and its bearer (figure 5.14.b), since the heart of a person is not an inseparable part of a person and, hence, their life spans can be completely independent. A heart can pre-exist its bearer as well as survive its death. Nonetheless, a heart must be part of a person, only not necessarily the same person in all possible circumstances. For these cases, of generic dependence from the part to a whole, we use the term *parts with mandatory wholes*.

Definition 5.16 (mandatory whole): An individual y is a mandatory whole for another individual x iff, x is generically dependent on a universal U that y instantiates, and x is, necessarily, part of an individual instantiating U :

$$(52). \text{MW}(U, x) =_{\text{def}} \Box(\mathcal{E}(x) \rightarrow (\exists U, y)(x < y)).$$

■

Once more, the distinction between inseparable parts and parts with mandatory wholes is neglected in practically all conceptual modeling languages. For this reason, we propose to extend the current UML aggregation notation with the Boolean meta-attribute ***inseparable*** to represent inseparable parts. When *inseparable* is equal to *true*, the minimum cardinality constraint in the association end corresponding to the whole universal must be at least 1. If *inseparable* is equal to *false*, the tagged value textual representation can be omitted. A UML class representing a whole

universal involved in an aggregation relation with minimum cardinality constraint of at least 1 in its association end represents a universal whose instances are mandatory wholes.

5.5 Part-Whole Theories in Linguistics and Cognitive Sciences

In this section, we start by reviewing perhaps the most well known linguistic study on the classification of part-whole relations. In an article entitled “*A taxonomy of part-whole relations*”, (Winston & Chaffin & Herrmann, 1987) (henceforth WCH), propose an account of the notion of part-whole by elaborating on different ways that parts can related to a whole. This study led to a refinement on the formal relation of *partOf* by distinguishing the six types of meronymic relations which are represented in table 5.1 below.

Table 5-1 Different types of Meronymic relations according to (Winston & Chaffin & Herrmann, 1987)

Relation	F	H	S	Examples
Component/ Functional Complex	+	-	+	handle-cup, punch line-joke
Member/Collection	-	-	+	tree-forest, card-deck
Portion/Mass	-	+	+	slice-pie, grain-salt
Stuff/Object	-	-	-	gin-martini, steel-bike
Feature/Activity	+	-	-	paying-shopping, dating-adolescence
Place/Area	-	+	-	Everglades-Florida, oasis-desert

The meta-properties that are used to create these distinctions, namely Functionality (F), Homeomerousity (H) and Separability (S), are explained as follows in a quote from the authors: “Functional parts are restricted by their function, in their spatial or temporal location. For example, the handle of a cup can only be placed in a limited number of positions if it is to function as a handle. Homeomerous parts are of the same kind of thing as their wholes, for example (slice-pie), while non-homeomerous parts are different from their wholes, for example (tree-forest). Separable parts can, in principle, be separated from the whole, for example (handle-cup), while inseparable parts cannot, for example (steel-bike).” (Winston & Chaffin & Herrmann, 1987).

This proposal makes an important contribution in acknowledging that there are different ways that parts composing a whole can relate to each other and to the whole they compose. Nonetheless, the study is overly linguistically motivated, focusing on the linguistic term “part-of” (and its

cognates) independently of the ontological and conceptual adequacy of the proposed distinctions.

Firstly, the notion of separability employed by the authors is very different from the one discussed in section 5.4.2 of this chapter. In their case, separability is based on whether “parts can/cannot be physically disconnected, in principle, from the whole to which they are connected.” (ibid.). The first problem with this definition is that it applies exclusively to physical objects, since it relies on the notion of “possible physical disconnection” (Gerlst & Pribbenow, 1995). It says nothing, for example, about whether the first act of *Turandot* is separable or not from the whole play. But more importantly, this notion of separability is of little use for conceptual modelling, since it does not elucidate anything about the ontological dependence relations between parts and wholes, their life-time dependencies, their identity and persistence conditions, etc. If a concept of separability that affords this type of analysis is adopted (such as the one advocated in section 5.4.2), many counterexamples can be found to question the authors’ classification. For instance, the relation between a human brain and a human body is of type Component/Functional Complex. However, under the assumptions we have made here, it is not the case that the brain is separable from the body. It certainly can be “physically disconnected”, however, due to the existential dependence between brain/body, in the case of separation it will cease to exist as a brain, and, since brain is a kind (in the sense of chapter 4), it will cease to exist as the same object. Therefore, one cannot state that it is the SAME object which has been separated from the body. Conversely, the authors deem the Stuff-Object example of a portion of steel that is part of a bicycle as an instance of an inseparable part. Once more, if an ontologically meaningful notion of separability is adopted, this example is no longer true: if one has a physical-chemical way of separating the steel which constitutes the bike from the bike itself (or the gin from the martini), there is no reason why one should not believe that it is the very same portion of steel (gin) that persists through the separation process. Moreover, the portion of steel might precede the creation as well as survive the destruction of the bike. Additionally, the bike could certainly be made of a different portion of steel. In summary, they are mutually independent from an existential point of view and, hence, separable.

The original taxonomy proposed by WCH has been refined by other authors. For instance, (Gerlst & Pribbenow, 1995) have noticed that the distinction between (a) Component/Functional Complex and (b) Feature/Activity is a superfluous one. What examples such as handle-cup and punch line-joke have in common is a certain role that the part plays w.r.t. the whole, and it is the functionality associated with the role that typically poses constraints on the internal (spatial or temporal) structure of

the whole. The same holds for dating-adolescence and paying-shopping. The only difference is that in the former case (a), the relata are endurants whilst in (b) they are perdurants³⁵.

(Gerlst & Pribbenow, 1995) as well as (Hornsby & Egenhofer, 1998) have noticed that the example of Place/Area are not in any important sense different from the cases of Component/Functional Complex and Feature/Activity aforementioned. Examples such as oasis-desert make clear that the part plays a role w.r.t. to the whole (e.g., providing water) and that, contra WCH, it does not constitute a case of homeomerousity. That is, the parts of desert are not of the same kind as the desert itself (an oasis is not a desert!). Likewise, the Everglades does not instantiate the same universals as Florida, since the latter possesses many properties which the former lacks (e.g., having a governor) and vice-versa.

(Gerlst & Pribbenow, 1995) have also observed that the examples of Stuff/Object used by WCH misconceive the phenomena it is supposed to represent. By attempting to differentiate between the linguistic expressions “be partly” (e.g., a martini is partly alcohol) and “made of” (e.g., the lens is made of glass), in the case of Stuff/Object parthood the authors have conflated two propositions: one which is indeed a part-of relation and one which represents a case of *constitution*. For instance, the sentences

- (i) A Martini is partly alcohol;
- (ii) A bike is partly steel

can be decomposed in two propositions

- (iii) A part of Martini is made of alcohol;
- (iv) A part of the bike is made of steel

In other words, instead of being primitive, this type of part-whole relation can be analyzed in terms of a constitution³⁶ relation and one of the other types of part-whole relations (mass/quantity in the case of (iii) and component/Integral Object in the case of (iv)).

In summary, the refinement of the WCH taxonomy proposed in (Gerlst & Pribbenow, 1995) contains the following categories:

- (a). Mass/Quantity.
- (b). Member/Collection;
- (c). Component/Functional Complex;

³⁵ see discussion of endurants and perdurants on chapter 6.

³⁶ Constitution is not considered here as an example of a part-whole relation. For instance, it does not satisfy any of the supplementation axioms (Masolo et al., 2003a).

These three new types of part-whole relations are discussed in the following subsections. The Mass/Quantity relation (a) is discussed in subsection 5.5.1; The relations of type (b) and (c) are discussed in subsection 5.5.2.

5.5.1 Quantities

One difference between Mass/Quantity and the other two types of parthood is that the relata of this relation always belong to the category of *amounts of matter* (masses, quantities), while in the Component/Functional Complex and Member/Collection they are *objects*. Quantities (such as water, sand, sugar, martini, wine, etc.) lack both *individuation* and *counting principles*. For this reason, the general terms which are linguistically represented by mass nouns (the linguistic counterpart of amounts of matter) cannot be used to substitute X, Y and Z in sentences such as:

- (i) (*exactly*) five X...;
- (ii) the Y which is Z.

A substitution for, for example, water in sentence (i) is not viable, since arbitrarily many parts of water are still water. Likewise, a success in the substitution by water in (ii) depends on the possibility of determining the referent and judge identity statements of individual quantities of water. What exactly should be that referent? Before answering this question we should call attention to what exactly is meant by *homeomerousity* and its relation to the WCH taxonomy.

Traditionally, homeomerousity means that an individual only has parts which are of the same kind (Zimmerman, 1995). This is clearly not the case for all amounts of matter, as the Gin-Martini case demonstrates. However, one can still say that every subquantity of Martini is again Martini and that although Martini is composed of Gin, Gin is itself “homeomerous” in this more liberal sense. This line of reasoning seems to suggest that homeomerousity is equated with infinite decomposability, i.e., for every subquantity of Martini there is always a subquantity of Martini, and the same holds for quantities of Gin. Some authors (e.g., Zimmerman, 1995), nonetheless, admit the existence of quantities of type K having K-atoms, i.e., individuals of type K that have no parts of the same type K. Examples include concrete mass terms such as ‘furniture’, ‘cutlery’ or ‘crowd’. These allegedly exemplars of quantities are definitely not homeomerous, not even in the more liberal sense. For example, there are parts of a crowd, namely individual persons, which is not are a crowd themselves and which are not

homeomerous in any meaningful sense³⁷. What can be said in this case is that these aggregates have a uniform structure and, in parity with (Gerlst & Pribbenow, 1995), we consider them as examples of *member/collection* parthood instead.

One could also consider homeomerousity to simply mean that an aggregate can merely have some parts of the same kind while having other parts of other kinds. However, if this were to be the case one could not use it as a meta-property to differentiate Mass/Quantity from Member/Collection. Notice that examples of “homeomerous” parts in this sense can be easily found for Member/Collection: a crowd can be part of a larger crowd; a forest can be part of larger forest.

Since WCH do not consider member/collection to be homeomerous, they would have to agree that quantities should be considered necessarily infinitely divisible in quantities of the same kind. We thus take the liberal approach and consider the case of real homeomerousity as a special case of infinitely divisible pure substances.

Now, an important question that comes to the mind is how we should represent in conceptual models the universals whose instances are quantities in the sense just mentioned? As discussed in chapter 4, individuals in a conceptual model must have a determinate identity, a requirement which is put forward since the initial conceptions of the first conceptual modeling languages. For instance, (Borgida, 1990) writes that, in both semantic data models and knowledge representations, “*an individual is assumed to have a unique, intrinsic identity*”. As we have discussed in chapter 4, in order to be able to make viable references to general terms which are not count nouns (mass terms, adjectives, verbs) they first must be nominalized. A nominalization of a mass noun, verb or an adjective promotes the shift to the category of count nouns (e.g., the fall of Jack, a lump of clay), hence, allowing for the representation of the corresponding sortal universals. An important question that then arises is: what is the best nominalization of mass terms so that they can be satisfactorily represented in conceptual models?

In order to investigate the possibilities let us take the example of a portion of wine which could be differentiated from other portions of wine by the year and source vineyard. What is the meaning (and implied principle of individuation) of a universal whose instances are portions of wine?

A first possibility is to consider the referent of the expression “the portion of wine” as a mereological fusion of all subportions of wine that

³⁷ Of course, one could say that a person is a physical object and that every part of a physical object is itself a physical object. However, if interpreted in this way, homeomerousity becomes completely non-informative since everything can be said to be homeomerous.

constitutes it. This approach is standard in philosophy and (Simons, 1987) suggests that quantities are probably the best case of application of the Classical Extensional Mereologies (CEM), since practically all objections raised against the CEM for the purpose of conceptual modeling can be safely lifted in the case of quantities and their parts. For instance, portions are always transitive and there is always a sum of two portions of quantity regardless how scattered they are. Nonetheless, and still from a philosophical point of view, the first problem with this conception of quantities is whether it is at all possible to have a principle of identity for portions of wine in this sense (Zimmerman, 1995; Lowe, 2001). A mereological principle of identity in this case prescribes that portion of wine A is equal to portion of wine B iff they have the same parts. However, since the parts of A and B are also portions of wine, to decide if A and B have the SAME parts one has to decide about the identity of the parts, and the parts of the parts, leading to an infinite regress, since, by assumption, quantities are infinitely divisible. One could derive some synchronic information about identity by saying that two quantities are different if they do not occupy the same region of space³⁸. However, this cannot be used as a diachronic principle of identity. Alternatively, one could say that a quantity A in t_1 is not the same quantity as B in t_2 if they have different properties such as volume or weight (still assuming the mereological principle). Nonetheless, the fact that B in t_2 has the same volume or weight as A in t_1 can only account for the sameness of the quantities in a very loose sense, meaning the same measure. In other words, in this case, the relation between A and B is one of equivalence, not one of *numerical identity*.

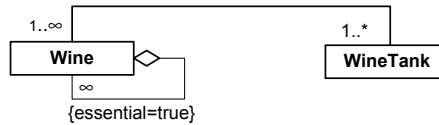
In conceptual modeling, there are a number of situations in which dealing only with qualitative identity of masses does not suffice. For instance, one might be interested in tracking the persistence of a quantity of a certain liquid which has been poisoned, or, in a chemical experiment, it could be important to track the change of properties in the very SAME persisting quantity. For this reason, contra (Gerlst & Pribbenow, 1995), who proposes that “quantities are arbitrary pieces of the whole as long as they are properly characterized by the quantitative measure”, we advocate that a treatment of masses in conceptual modeling must deal explicitly with the case of numerical identity.

However, there is still a bigger problem with this idea from a conceptual modeling stance. Figure 5.19 depicts the representation of a portion of Wine universal in the sense just mentioned. In this specification, the idea is to represent a certain portion of wine as the mereological sum of all subportions of wine belonging to a certain vintage. As it can be noticed,

³⁸ There is still the issue that space itself could be considered to be infinitely divisible (Lowe, 2001).

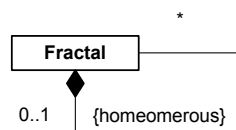
since every portion of wine is composed of subportions of wine, the cardinality of the part-whole relation cannot be specified in a finite manner. The same holds for every cardinality constraint for associations involving portions of wine.

Figure 5-19 Problems with the representation of Quantities as Mereological Sums



Furthermore, “homeomerous” entities represented in this manner can induce to representation errors in the presence of other shareability constraints. For example, figure 5.20 presents an exact copy of a UML class diagram from (Saksena et. al, 1998)³⁹ that symbolizes a Fractal (perhaps the prototypical example of homeomerous form). The intention of the authors seems to be to represent that a fractal, i.e., the rendering of one iteration step of an IFS, is part of only one instance of the infinite recursion of this function (Peitgen & Jurgens & Saupe, 1992). In other words, a part of an instantiation of a *Julia Series* is not a part of another instantiation of the same fractal form, or part of an instantiation of the *Mandelbrot Series*. However, that is not what is represented in the model. The model states that every instance of a fractal (i.e., every iteration of an IFS) is part of only one other fractal. This is clearly mistaken: the n^{th} iteration of an IFS is part of all previous iterations of the same fractal. This problem is far from being specific to Fractals. In fact, for all homeomerous entities, with the exception of the maximal sum of subquantities, all other parts of quantity are necessarily part of innumerable other quantities of the same kind.

Figure 5-20 Mistaken representation of homeomerous part with non-shareability (from Saksena et al., 1998)

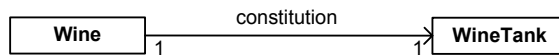


There is still one further philosophical argument invalidating this modeling alternative for the purpose of conceptual modeling. According to (Wandinger, 1998), masses are *identificationally dependent* of substantials that are instances of sortal universals. As he put it: “The formal concepts ‘amount’, ‘part’ and ‘stuff’, like ‘mass’ and ‘matter’, are formed ‘on the back of’ (Simons 1987, p. 191) the formal concept ‘(material) thing’ or ‘(material) object’. There is no mass, except the mass of a certain object.

³⁹ The tagged value {homeomerous} in the picture is a proposal of (Saksena et al., 1998) and it is present in the original article.

There is no stuff except the stuff a certain thing consists of". This is to say that individuating a quantity depends on the definite descriptions used for referring, which succeeds only when the referent can be individuated by sortals, i.e., it is the water in the bathtub, the clay that constitutes the statues, a cube of sugar, that can be referred to, not just some water, some clay, or some sugar. Moreover, quantities have no criteria for when they constitute a whole of some sort (unity criteria), except in cases in which we derive those criteria from objects that are only identifiable via sortals. This view is also supported in (Quine, 1960), who proposes that every occurrence of a quantity expression having the form "The K", "The same K" (where K is a concrete quantity universal) is really a masked reference to a *portion* of K to which an ordinary sortal universal applies. Alternatively formulating, as Quine puts it: "[in these situations, always] some special individuation standard is understood from the circumstances". This perspective gives rise to a second option of representation for the wine/wine tank example, as depicted in figure 5.21.

Figure 5-21
Representation of
Quantities as
Identifiably
Dependent of Objects



There are a number of observations that can be made about figure 5.21. In this second option for nominalization of quantities, Wine means the maximal content of a Wine Tank. Likewise, the referent of a portion of clay means whatever quantity of clay constitutes a given statue, which is in turn individuated by the principle of identity supplied by the sortal statue. In this representation, there is no longer a problem for the specification of cardinality constraints between portion of wine and wine tank: every wine tank has as its content one single definite portion of wine. Additionally, since wine portion means the maximal content of a wine tank, it is not the case that this concept is homeomeric, i.e., there is no part of a portion of wine which is itself a portion of wine (otherwise, it would not be the maximal content). Thus, there is no problem with infinite cardinalities, infinite divisibility and infinite domain of individuals (see section 5.1). Moreover, portion of Wine becomes a genuine sortal: it is always determinate if two portions of wine are identical and it is always determinate how many portions of wine there are. We emphasize that it is still convenient to consider portions of wine as having all parts as essential, i.e., if a part from a portion of wine is removed then it becomes a different portion of wine. The reason for this is that, otherwise, in some cases, the portion of wine becomes identical with the object it constitutes. Take the statue/lump of clay example. If A is the same lump of clay as long as it constitutes the same statue B, A would have necessarily the same properties as B and have a complete life-time dependency. For instance: (a) if a piece

of B is removed, B is still the same statue and so is A still the same lump of clay, since it still constitutes the same statue; (b) If the form of B is altered, B ceases to exist and so does A, since it no longer constitutes the same statue.

In summary, in this second alternative nominalization, a quantity is an *inseparable constituent* of the object it constitutes. In a manner which is analogous to the way such objects named *Features* in DOLCE (Masolo et. al, 2003a) (e.g., holes, bumps, edges, stains) are considered here *inseparable parts* of their hosts (see section 5.4.2).

Although this second alternative contains important advantages over the first one from a conceptual modeling point of view, it leads to a problematic consequence. The problem is implied exactly by the *rigid specific dependence* relation between a quantity in this sense and its container. As put by (Cartwright, 1965), a sentence such as the “same K” (where K is a quantity universal) should be understood in a such way that

x is the same K as y iff x is some K, y is some K, and (x = y)

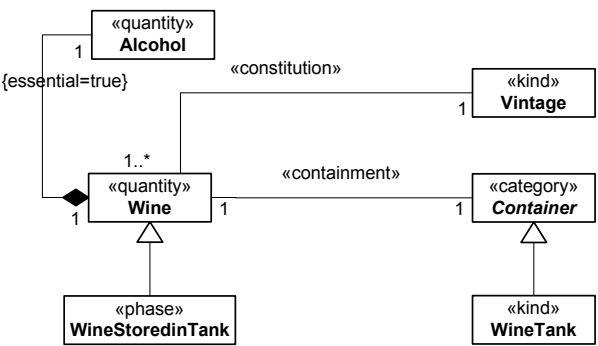
or, as discussed in chapter 4, in statements of identity, the relata must instantiate the same kind, i.e., the same rigid sortal supplying their principle of identity. In parity with (Cartwright, 1965; Zimmerman, 1995), we consider as meaningful a sentence such as “the sugar that was in that cube is the same sugar as the one in this lump”. However, if this is the case, which kind of individuation and identity principle should be applied to x and y, that of cubes or of lumps? Notice that x (or y) is not necessarily none of the above (in the modal sense) and there is not an ultimate substance sortal that would cover both universals.

A third nominalization alternative that solves this problem is presented as follows. This last option relies on the notion of *piece* discussed in (Lowe, 2001). According to Lowe, a piece of a quantity K is a maximally self-connected object constituted by portions of K (portions in the first sense discussed above). Like in the second nominalization alternative, a quantity of K in this sense is an instance of a genuine sortal universal, i.e., it has definite individuation, identity and counting principles. Moreover, it is not homeomeric, however it can still be composed of other quantities K' in the same sense of quantity (see figure 5.22). All its parts are also essential, and it does not contain the infinite regress problems mentioned for the first case. Nonetheless, differently from the second alternative, the dependence relation between a quantity and its container is a generic not a specific one. For this reason we can state that for the same maximally self-connected quantity of wine, there can be several “container phases”. This idea is represented in the (incomplete) model of figure 5.22. A vintage is an object constituted by (possibly many) quantities of wine, it is, however, not a

quantity since it can be scattered over many quantities. Moreover, it is not necessary for its constituent quantities to be essential: even if the quantity of wine now stored in a certain tank is destroyed, we still have numerically the same vintage. We, therefore, propose the use of this third alternative for the nominalization of quantities and their representation in conceptual models. From now on, we shall use the term *quantity of matter K* or *objectified portion of matter K* to refer to a *piece of K* in the Lowe’s sense aforementioned, and use the stereotype «quantity» to symbolize a kind whose instances are quantities in this sense.

We can summarize many points of the argument carried out in this section by using an example proposed by (Cartwright, 1965). If a sentence such as “Heraclitus bathed in some water yesterday and bathed in the same water today” is true then for some suitable substituends of *x* and *y* we have that: (a) *x* is a quantity of water and Heraclitus bathed in *x* yesterday; (b) *y* is a quantity of water and Heraclitus bathed in *y* today; (c) *x* = *y*. However, (c) when interpreted as (d) “the water Heraclitus bathed in yesterday = the water Heraclitus bathed in today” requires that: (e) there is exactly one *x* such that *x* is a quantity of water and Heraclitus bathed in *x* both yesterday and today. But, if quantity is interpreted in the first sense above, there are not one but infinitely many particulars that would satisfy (a), (b) and (d) without satisfying (c). The same does not hold for the second and third senses.

Figure 5-22
Representation of quantities as *maximally-self-connected portions* which are *generically dependent* of the continuants they constitute

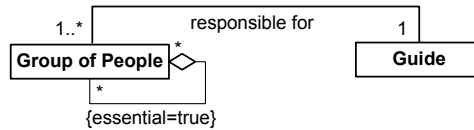


5.5.2 Collections and Functional Complexes

As recognized by WCH, collections such as tree-forest, card-deck, brick-pile, lion-pack cannot be said to be homeomerous wholes. Collections have parts that are not of the same kind (e.g., a tree is not forest) and they are not infinitely divisible. As a consequence, a representation of a collection as a mereological sum does not lead to the same complications as those encountered in the first alternative for the representation of quantities.

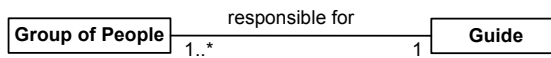
Take, for instance, the example depicted in figure 5.23, which represents a situation analogous to that of figure 5.19. Differently from the case with quantities in figure 5.19, there is no longer the danger of an infinite regress or the impossibility for specifying finite cardinality constraints.

Figure 5-23
Representations of
Collections with parts of
the same type



In figure 5.23, the usual maximum cardinality of “many” can be used to express that group of people has as parts possibly many other groups of people and that a guide is responsible for possibly many groups of people. Nonetheless, in many examples (such as this one), this model implies a somewhat counterintuitive reading. In general, the intended idea is to express that, for instance, John as a guide, is responsible for the group formed by {Paul, Marc, Lisa} and for the other group formed by {Richard, Tom}. The intention is not to express that John is responsible for the groups {Paul, Marc, Lisa}, {Paul, Marc}, {Marc, Lisa}, {Paul, Lisa}, and {Richard, Tom}. A simple solution to this problem is to consider groups of people as maximal sums, i.e., groups that are not parts of any other groups. In this case, depicted in figure 5.24, the cardinality constraints acquire a different meaning and it is no longer possible to say that a group of people is composed of other groups of people.

Figure 5-24
Representation of
Collections as Maximal
Sums



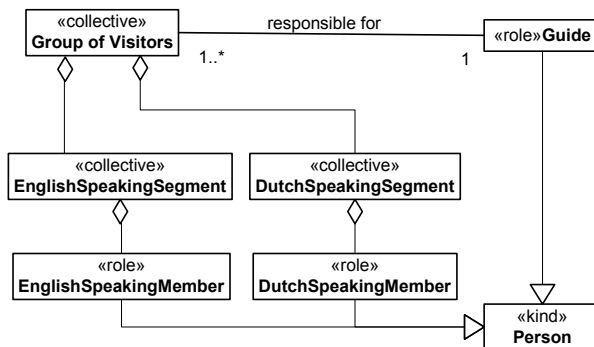
This solution is similar to taking the meaning of a quantity K to be that of a maximally-self-connected-portion of K . However, in the case of collections, topological connection cannot be used as a *unifying relation* (see section 5.3) to form an integral whole, since collections can easily be spatially scattered. Nonetheless, another type of connection (e.g., social) should always be found. A question begging issue at this point is: why does it seem to be conceptually relevant to find *connection relations* leading to (maximal) collections? We defend the idea that, alike quantities, arbitrary sums of collections make little cognitive sense. In the case of quantities, we are not interested in the sum of a subportion of my bathtub’s water with a subportion of the North Sea. Instead, we are interested in the quantity of water in that specific bottle, or lying in that specific location. Likewise, we are not interested in arbitrary collections of individuals but aggregations that have a purpose for some cognitive task. As discussed in section 5.3, integral wholes exist because there is some unifying relation that holds

among the totality of its parts and only among them, i.e., a closure system. For example, a group of people of interest can be composed by all those people that are attending a certain museum exhibition at a certain time. Now, by definition, a closure system is maximal (see definition 5.4), thus, there can be no group of people in this same sense that is part of another group of people (i.e., another integral whole unified by the same relation).

Nonetheless, it can be the case that, among the parts of a group of people, further structure is obtained by the presence of other collections unified by different relations. For example, it can be the case that among the parts of a group of people A, there are collections B and C composed of the English and Dutch speakers, respectively. This situation is depicted in figure 5.25. Neither the English speaking segment nor the Dutch speaking segment is a *group of people* in the technical sense just defined, since a group of people has properties that apply to none of them (e.g., the property of having both English and Dutch segments). Moreover, the unifying relations of B and C are both specializations of A's unifying relation. For example, A is the collection of all parties attending an exhibition and the B is the collection of all English speakers among the parties attending the same exhibition. For this reason, transitivity holds unrestricted among the part-whole relations depicted in figure 5.25.

We shall use the stereotype «collective» to symbolize a kind whose instances are collections.

Figure 5-25
Representation of
Collections



Judging from this perspective, collections bear a strong similarity to functional complexes in the classifications of WCH and Gerlst & Pribbenow. This similarity is in fact acknowledged by (Gerlst & Pribbenow, 1995) in the following passage: “Depending on what aspect of this structure [i.e., the thing’s structure] one focuses on, different types of part-whole relations are possible. However, in many cases there is a primary or prototypical view...the primary view critically depends on the granularity

level which is assumed for classifying an entity...configurations such as ‘a pile of books’ are on one hand similar to collections and on the other hand similar to complexes”. In the course of the article, the authors propose that the difference between a collection and a functional complex is that whilst the former has a *uniform* structure, the latter has a *heterogeneous* and *complex* one.

We propose to rephrase this statement in other terms. In a collection, all member parts play the same role type. For example, all trees in a forest can be said to play the role of a forest member. However, a tree is not necessarily a forest-part, i.e., the latter is an anti-rigid concept, and representing the part-whole relation between forests and trees via the role type *forest-tree* prevents one from having to specify minimum cardinality relations which are zero. In complexes, conversely, a variety of roles can be played by different components. For example, if all ships of a fleet are conceptualized as playing solely the role of “member of a fleet” then it can be said to be a collection. Contrariwise, if this role is further specialized in “leading ship”, “defense ship”, “storage ship” and so forth, the fleet must be conceived as a functional complex. In summary, collections as *integral wholes* (i.e., in a sense that appeals to cognition and common sense conceptual tasks) can be seen as limit cases of Gerlst & Pribbenow’s *functional complex*, in which parts play one single role forming a uniform structure.

Finally, we emphasize that, differently from *quantities*, collectives do not necessarily obey an extensional principle of identity. Some collectives can be considered extensional by certain conceptualizations. In this case, the addition or subtraction of a member renders a different collective. However, we also consider here the existence of *intentional collectives* obeying non-extensional principles of identity (Botazzi et al., 2004).

5.6 The Problem of Transitivity Revisited

In section 5.3, we have addressed the problem of fallacious transitivity inferences in part-whole relations by employing the notion of contexts. However, at that point, we are still dealing with one single type of parthood inherited from the formal part-whole relation in mereology. A question begging issue at this point is: by taking advantage of the refinement of this relation developed in section 5.5, can we provide more direct guidelines for prescribing or proscribing transitive relations? In other words, can we establish some relations between the specific type of part-whole relation and the issue of transitivity?

The distinction between the three types of part-whole relation discussed in the previous section (quantity/mass, member-collection,

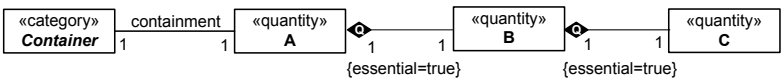
component-functional complex), actually reflects a distinction among different types of relata.

Mass-Quantity:

The quantity/mass relation holds between *quantities* (in the technical sense explained in section 5.5.1). Let us suppose a model such as the one depicted in figure 5.26, in which A, B and C are quantities. We can show that for any A, B, C, the part-whole relation ($C < A$) holds as a result from the transitivity ($C < B$) and ($B < A$). The argumentation can be developed as follows: if A is a quantity then it is a maximal portion of matter unified by the characteristic relation of self-connectedness. That is, any part of A is connected to any other part of A. If B is part of A then B is connected to all parts of A. Likewise, if C is part of B then C is connected to all parts of B. Since connection is transitive, then we have that C is connected to all parts of A. Thus, since A is unified under self-connection, C must be part of A (otherwise the composition of A would not be a closure system, see definition 5.4). Therefore, we conclude that for the case of quantities, transitivity always holds.

Another way to examine this situation is by inspecting A at an arbitrary time instant t. We can say that all parts of A are the quantities that are contained in a certain region of space R (i.e. a topoid, see Guizzardi & Herre & Wagner, 2002a). Since A is an objectified matter, than the topoid R occupied by A must be self-connected. Therefore, if B is part of A then B must occupy a sub-region R', which is part of R. Likewise, if C is part of B, it occupies a region R'', part of R'. Since spatial part-whole relations are always transitive (Johansson, 2004), we have that R'' is part of R, and if C occupies R'', then it is contained in R. Ergo, by definition, C is a part of A.

Figure 5-26 Part-Whole relations among quantities



Parthood relationships between quantities are always non-shareable. For instance, in figure 5.23, B can only be part of one single quantity of A, since A is a maximal. Moreover, A has at maximum one quantity of B as part, since B is also a maximal portion. Finally, as discussed in section 5.5.1, every part B of A is essential.

As in figure 5.26, we decorate the standard UML symbol for composition with a Q to represent a quantity/mass parthood relation. If cardinality constraints are fully specified, then the Q-parthood is a relation which:

- (i) is non-shareable;
- (ii) the part is essential to the whole;
- (iii) the cardinality constraints in both association ends are of one and exactly one;
- (iv) only holds between quantities;
- (v) it is transitive, i.e., for all a, b, c , if $Q(a, b)$ and $Q(b, c)$ then $Q(a, c)$.

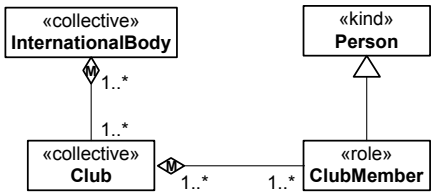
The second type of parthood considered here is the *member-collection* relation. Notice that the relation between *EnglishSpeakingSegment* and *Group of People* in figure 5.25 does not exemplify this type of relation. Instead, it is a relation of type *subcollection-collection*. The difference between the two types of parthood can be linguistically motivated in the following manner. According to (Vieu & Aurnague, 2005), classical semantic analysis of plurals and groups (Landman, 1996; Link, 1991) distinguish between atomic entities, which can be singular or collectives, and plural entities. Any collective atomic entity (e.g., a group of men, the herd) is constituted of some plural entity (e.g., {John, Paul, George, Ringo}, or the cows, which are sum of man-atoms, or cow-atoms, respectively). Following (Aurnague & Vieu, 2005), we have that:

Member-Collection:

The *member-collection* relation is one that holds between a *singular entity* and either a *plural* or a *collective term*. This is the relation, for example, between John and a group of men, the cow Joanne and the herd, or between the province of Overijssel and The Netherlands. Member collection relations are never transitive, i.e., they are intransitive. This can be understood in our analysis in the following way. To say that a member must be a *singular* entity coincides in this case with this entity being an *atom of a given context*. Or to put it differently, the unifying relation underlying membership cannot be further refined. An example of a member-collection relation is the following case from (Henderson-Sellers & Barbier, 1999): “I am member of a club (collection) and my club is a member of an International body (collection). However, it does not follow that I am a member of this International body since this only has clubs as members, not individuals”. This situation is depicted in figure 5.27 below, in which we decorate the standard UML symbol for aggregation with an M to represent a member/collection parthood relation. A M-parthood relation is intransitive, i.e., for all a, b, c , if $M(a, b)$ and $M(b, c)$ then $\neg M(a, c)$. Therefore, in this model, although the *ClubMember* John is part of the *F.C.Twente Club*, and that the *F.C.Twente* is part of the an international body Ψ , John is not part of (member of) Ψ .

The statement above about the intransitivity of the M-parthood can be made more general. Members of a collection are considered to be atomic w.r.t. the context in which the collection is defined. As a consequence, if an individual x is a part of (member of) a collection y , then for every z which is part of (memberOf, componentOf, subCollectionOf) x , z is not a part of (member of) y . Thus, for instance, although an individual John can be part of (member of) a Club, none of John's parts (e.g., his heart) is part of (member of) that Club.

Figure 5-27 Examples of member/collection part-whole relations



Subcollection/Collection:

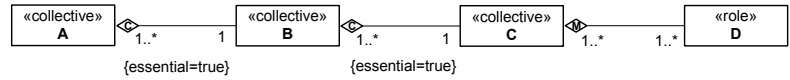
The *subcollection-collection*, conversely, is a relation that holds between two plural entities, or collectives constituted by such plural entities, such that all atoms of the first are also atoms of the second. In our analysis, subcollections are always created by refining the unifying relation of a certain collection. In the example of figure 5.25, the *EnglishSpeakingSegment* is part of the *Group of Visitors*. The latter is unified by, for instance, taking all people that visit an exhibition x at date y . The former by taking all people that visit an exhibition x at date y , and that speak English. As another example, take a forest and the north part of the forest. The latter is a part of (subcollection-of) the former. The forest is unified by taking all the trees located in a given area A . The north-part of the forest is unified by the taking all the trees located in the north part of A .

As in figure 5.26, we decorate the standard UML symbol for composition with a C to represent the subcollection/collection parthood relation. If cardinality constraints are fully specified, then the C -parthood is such that:

- (i) the part is essential to the whole;
- (ii) the cardinality constraints in the association end relative to the part is one and exactly one;
- (iii) only holds between collectives;
- (iv) it is transitive, i.e., for all a,b,c , if $C(a,b)$ and $C(b,c)$ then $C(a,c)$. In this figure this means that every C is also part of an A .

Notice that if we have $M(x,y)$ and $C(y,z)$ then it is also the case that $M(x,z)$, since this would clearly be a case of $(x <_A y)$ and $(y <_B z)$, where if R' and R are the unifying relations of A and B then R' is a refinement of R . Therefore, in figure 5.28 we have that every D is a part of (member of) a B and also a member of an A . In figure 5.25, this means that every *EnglishSpeakingMember* is part of (member of) the *Group of Visitors*.

Figure 5-28 Examples of SubCollection/Collection and Member/Collection part-whole relations



Component-Functional Complex:

Now, we can turn our attention to the most important type of parthood relation for the purposes of conceptual modeling, namely, the component/functional complex relation.

Differently from collectives, complexes are composed by parts that play a multitude of roles in the context of the whole. As a consequence, in practical cases, the isolation of contexts for complexes can result in a complicated task, and in particular, one that requires a strong knowledge of the domain being modeled. We believe it is reasonable to assume that conceptual modelers working on the representations of specific domains have access to sufficient domain knowledge that enables them to define unifying relations for complexes, and to consequently isolate suitable contexts. Nonetheless, as we show in the remaining of this section and also in section 7.4, we can provide some patterns of relations for the case of component/functional complex parthood, in which transitivity is guaranteed to hold. The proposal of these patterns is very much in line with one of the main objective of this thesis, namely, to provide well-founded engineering tools for conceptual modelers.

The parts of a complex have in common that they all possess a functional link with the complex. In other words, they all contribute to the functionality (or the behavior) of the complex. Therefore, if it is generally the case that essential parthood entails dependence (see definition 5.11), in this type of parthood relation, an essential part represents a case of *functional dependence*. To put it more precisely, for all complexes, if x is an essential part of y then y is functionally dependent on x .

Let us take the example depicted in figure 5.29⁴⁰. According to definition 5.11, we have that:

⁴⁰ Since the component/complex parthood relation is the most common one in conceptual modeling, we use the standard presentation of the UML aggregation symbol to represent relations of this type. In the examples concerning these relations in the remaining of this section, for simplification, we shall refrain from specifying cardinality constraints.

- (i) $\Box(\forall \text{Person}, x)(\exists! \text{Brain}, y) \Box(\mathcal{E}(x) \rightarrow (y < x));$
(ii) $\Box(\forall \text{Brain}, x)(\exists! \text{Cerebellum}, y) \Box(\mathcal{E}(x) \rightarrow (y < x)).$

From (i) and (ii) it follows that

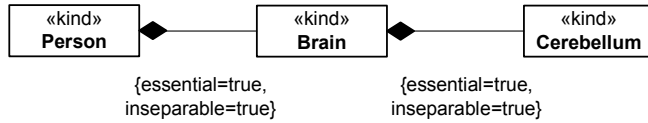
- (iii) $\Box((\forall \text{Person}, x)(\exists! \text{Cerebellum}, y) \Box(\mathcal{E}(x) \rightarrow \mathcal{E}(y))).$

Hence, following definition 5.10, we have that

- (iv) $\Box((\forall \text{Person}, x)(\exists! \text{Cerebellum}, y) \text{ed}(x, y)),$

i.e., every person is dependent on a specific cerebellum.

Figure 5-29 Examples of (functional) essential parthood between complexes



In chapter 6, we use the notion of existential dependence (definition 5.10), to distinguish between what we name *substantial* and *moments*. In short, substantials are concrete objects of every day experience such as cats, persons, houses, cars, brains and cerebellums. Moments, in contrast, are particulars such as a weight, a thought, an electric charge, a smile, a hand shake. Substantials are mutually independent, except in case they are related via parthood, i.e., in the case of essential or inseparable parts. That is to say that if a substantial x is external to a substantial y (i.e., disjoint from y) then y is independent of x (see formula 10 in chapter 6).

Now, since the elements in figure 5.29 are substantials, and since we have shown that every person is dependent of a specific cerebellum, if the cerebellum x is not part of the person y , we have to conclude that y is existentially dependent on a substantial that is disjoint from it, and consequently, that y (a particular person) is not a substantial. This result is certainly absurd. Therefore, we must conclude that transitivity always holds across essential parthood relations.

Figure 5.30 is an extension of the model of figure 5.29 by adding the role universal Student. We can show that, in this case, transitivity holds between the relationships cerebellum/brain and the inherited brain/student (represented by a dashed line) in the figure. Since the semantics we assume for the subsumption relation is that $\Box((\forall \text{Student}, x) \text{person}(x))$ (see section 4.1), from this and from (i) and (ii) above we can show that

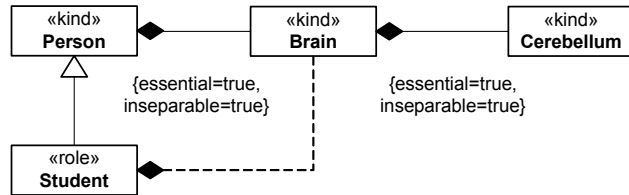
- (v) $\Box((\forall \text{Student}, x)(\exists! \text{Cerebellum}, y) \Box(\mathcal{E}(x) \rightarrow \mathcal{E}(y))),$

or equivalently, that

$$(vi) \quad \square((\forall \text{ Student}, x)(\exists! \text{ Cerebellum}, y) \text{ ed}(x, y)).$$

Now, since a student is also a substantial then we must conclude that every student has a cerebellum as an essential part.

Figure 5-30 Transitivity of parts in the case of Roles



In section 7.4, we resume this discussion to consider also the case of *generic (functional) dependence* between complexes, i.e., in which situations transitivity holds across relationships of *mandatory* (as opposed to *essential*) *parthood* between functional complexes.

5.7 Related Work

In this section, we discuss several attempts in the literature to provide a theoretical foundation for part-whole relations in conceptual and object-oriented modeling.

5.7.1 The Approach of Odell and Bock

In an article entitled “*Six Different Kinds of Composition*” (Odell, 1998), James Odell has proposed an adaptation of the taxonomy of part-whole relations proposed by Winston, Chaffin and Herrman (see section 5.5) for the purpose of modeling object-oriented systems. We recognize the pioneering nature of Odell’s work as one the first attempts in the object modeling literature to emphasize the multitude of part-whole relations that should be considered. However, there are a number of important issues in which our approach differs from his, some of which are related to features that Odell’s approach inherits from WCH.

On page 3 of the “*Six Different Kinds of Composition*”, the author makes the following comment on, what he terms, *material-object* composition (stuff-object in WCH’s terminology): “The word *partly* is not a requirement of the material-object relationship. For instance, a windshield could be made entirely of glass-not just partly. Other material-object relationships

require a subjective judgment. For example, can the ceramics (of the spark plugs) be removed from a car? If so, ceramics is a component-integral object relationship, instead". From our perspective, in this passage there is confusing between the relations of parthood and constitution. If a glass would be considered the only part of a window, it would not satisfy the weak supplementation axiom, which is considered a minimum requirement for all part-whole relations (Simons, 1987). However, the biggest problem is with the second part of the statement: the difference between a spark plug and the ceramics it is made of is not a subjective one, but an ontological one. Akin to the statue/lump of clay problem, the plug and the quantity of ceramics in its *constitution* have different modal properties and, by Leibniz law, cannot be deemed identical. Finally, if we were to respect our common sense intuitions, both the ceramics and the spark plugs should be considered separable parts of the car. In other words, if separability is taken in an ontologically meaningful modal sense (see section 5.5): the ceramics (or the plugs) pre-existed the car, i.e., there are worlds in which the ceramics (or the plugs) exist without being part of the car.

This conception of separability (as being physically entangled) has its source in the original WCH proposal. However, they reappear in other parts of Odell's article. For instance, on page 4 of the same paper, he proposes that the difference between a place-area and portion-object compositions (portion-mass in WCH's terminology) is that while both are homeomerous, only the former is constituted solely by inseparable parts. As discussed in section 5.5, place-area compositions cannot be deemed homeomerous in any meaningful sense of the word, but, other than that, it is not the case that their parts are necessarily inseparable either. For instance, the province of Trentino-Alto Adige is a (place-area) part of Italy, but not an inseparable part, since there are worlds (namely before 1921), in which it belonged to the Austrian-Hungarian Empire. This is far from being an isolated case: Uruguay used to be part of Brazil (under the name of Cisplatina), Strasbourg used to be part of Germany, and so forth. In summary, there is nothing in the place-area composition relation that requires the parts to be inseparable.

One of the main intentions of the typology of meronymic relations proposed by Winston, Chaffin and Herrmann (WCH) is to address the problem of fallacious cases of transitivity in part-whole relations. By adopting the WCH framework, Odell states that "when the same kind of [part-whole] relationship is used, the conclusion is always correct". A similar statement is also found in (Artale et al., 1996): "as long as we are careful to keep a single sense of part, it seems that the part-whole relation is always transitive. However, when we inadvertently mix different meronymic relations problems with transitivity arise". As we discussed in section 5.6, *member-collection* relations are intransitive and *component-functional complex*

relations are non-transitive (i.e., transitive in certain circumstances and not transitive in others) irrespective of whether we maintain the same “sense of part”.

As we have also discussed in section 5.5.1, if homeomerousity is given its standard interpretation in philosophy, portion-mass compositions are not homeomerous either. This actually becomes clear in an alleged example of portion-mass composition mentioned by Odell (p.3, *ibid.*): “*A meter is part of a kilometer*”. Instances of Meter and Kilometer clearly do not have the same properties (e.g., only the latter has the property of corresponding to 1000 meters). Although both are subtypes of Measure, this should not be an argument for considering it a case of homeomerousity. After all, any part of a physical object can be considered to be a physical object.

In differentiating the member-bunch composition (WCH’s member-collection) from the component-functional complex, Odell writes (page 4, *ibid.*): “In the composition relationships above, the parts bear a particular functional or structural relationship to one another or to the object they comprise. Member-bunch composition has no such requirement...The member-bunch relationship is different-based, instead, on spatial proximity or social connection. For a shrub to be part of a garden implies a location close to the other plants. For an employee to be part of a union implies a social connection”. As we have discussed in section 5.3, in order to provide a suitable foundation for conceptual part-whole relations, a mereology (a theory of parts) should be complemented with a theory of wholes. This is because the aggregates that interest us in a commonsense ontology of reality are integral entities, unified by genuine relations, not arbitrary formal ones. In fact, in our theory, an integral whole is defined in terms of closure systems containing all and only those objects that bear a unifying (material) relation to one another. Now, if this is the case, this should hold for all types of conceptual aggregates, i.e., both for forests and televisions. Otherwise, if trees in a forest bear no genuine relation to each other, how can we decide which trees are part of the forest and which are not? The point we want to emphasize here is that the relationships unifying collectives are genuine unifying relations, and that deciding whether they are “functional or structural” may be a matter of conceptualization. After all, for instance, employees who are *socially connected* to each other can very well be functionally related to each other.

Finally, in section 5.5.2, we have articulated a distinction between member-bunch and component-functional complex based on the fact that only in the former all parts seem to play the same role w.r.t. to the whole. In the examples of *member-partnership* (a special case of member-bunch) provided by Odell (e.g. “Ginger and Fred are a waltz couple”, p.5), the members can only be said to play *the same role* if one is willing to raise to a certain extent the level of abstraction. This seems to support our argument

that perhaps the distinction between member-collection and component-integral object is strongly dependent on the conceptualization at hand.

5.7.2 The BWW Approach

In (Opdahl & Henderson-Sellers & Barbier, 2001), the authors employ the BWW (Bunge-Wand-Weber) ontology (Wand & Weber, 1993; Weber, 1997) as a foundation for a conceptual framework defining a taxonomy of part-whole relations in terms of their *primary* (e.g., reflexivity, asymmetry and transitivity), *secondary* (e.g., shareability, mutability, separability) and *consequent properties* (e.g., ownership, propagation of operations, encapsulation). Moreover, the article analyses the different kinds of part-whole relations in terms of ontological soundness, i.e., if the proposed concepts are meaningful in terms of real-world semantics. Finally, some UML stereotypes are proposed in order to provide syntactical representations to the proposed ontological distinctions.

Here we are only interested in those properties that are meaningful in terms of ontological correspondence. For that reason, we have chosen to ignore properties that are relevant only in terms of implementation decisions (e.g., ownership, propagation of operations, encapsulation, by-value or by-reference, used or not used).

In terms of the primary and secondary characteristics of part-whole relations there are three points of disagreement between our proposal and the one of (Opdahl & Henderson-Sellers & Barbier, 2001):

Emergent and Resultant properties

Both (Opdahl & Henderson-Sellers & Barbier, 2001) and (Wand & Storey & Weber, 1999) propose that we should only model a *thing* as an aggregate if we are interested in modeling its emergent and resultant properties. Emergent and Resultant properties are defined by Bunge in (1977, p.97) as follows: “P is a *resultant* or *hereditary property* of x iff P is a property of some component $y \in C(x)$ (see definition 5.1) other than x; otherwise P is an *emergent* or *gestalt property* of x”.

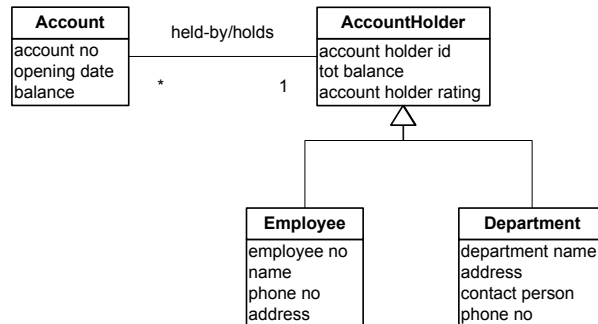
The UML class diagram depicted in figure 5.31 is an exact copy from (Opdahl & Henderson-Sellers & Barbier, 2001, p.391). In describing this model, the authors explicitly state that in the problem domain all *Departments* are aggregates of *Employees*. Nevertheless, Opdahl and colleagues deem unacceptable to add a part-whole between these two classes since the diagram would comprise no resultant/emergent property of Department relative to Employee. We strongly disagree with this view and we think that this restriction arises from a misinterpretation of Bunge’s ontology.

According to Bunge, every aggregate certainly has emergent and

resultant properties. However, his ontology makes explicit the distinction between the properties possessed by a thing and the representations of these properties, namely attributes. According to Bunge, there are no bare individuals, i.e., things without properties: a thing possesses at least one substantial property, even if we humans are not or cannot be aware of them. Humans get in contact with the properties of things exclusively via the things attributes, i.e. via a chosen representational view of its properties.

In summary, we agree that emergent/resultant properties are basic characteristics of part-whole relations, in the sense they are present in all of them. What we do not agree is to use the existence of resultant/emergent *attributes* as a criterion for deciding whether to represent part-whole relationships. In other words, emergent/resultant properties of aggregates will always exist but we do not have always to be interested in them and sometimes we cannot even be aware of them. Additionally, we think that the representation of these attributes is not a necessary condition for one to benefit from the representation of part-whole relations in terms of communicability, understanding and problem-solving.

Figure 5-31 A part-whole relationship between Employee and Department belonging to the domain is not represented in the model because neither *emergent* nor *resultant* "properties" are represented (from Opdahl & Henderson-Sellers & Barbier, 2001, p. 391)



Mixing-up different properties

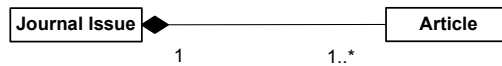
Lifetime dependency is a characteristic of part-whole relations with essential and/or inseparable parts. In this sense, we disagree with examples such as the one used in (Henderson-Sellers & Barbier, 1999) to justify the existence of parts that are separable, but that share the same destruction as the whole: "a car wheel is independent of the car but if the wheel is in the car during the car's destruction then it is also destroyed". In this case, the wheel is clearly separable from the car, it just happened to be the same event that caused the destruction of both objects (had the wheel been separated from the car, the car's destruction would not propagate to the wheel).

This confusion seems to be motivated by an object-oriented programming bias towards conceptual modeling. Traditionally, in OO programming languages, an object can be made responsible for the destruction of other objects as a procedure for memory deallocation named *garbage collecting*. Thus, it can be warranted that an object X should trigger the destruction of other objects coupled with X in the moment of its destruction, even if the coupling is merely a contingent one.

The mixing of different conceptual properties due to an implementation-oriented attitude is not uncommon in conceptual modeling. Another example that is recurrent is the unfounded association between shareability and separability. For instance, (Saksena & France & Larrondo-Petrie, 1998) claim that “shared parts are necessarily separable”, and that “sharing implies that if the membership of a part in one aggregate is nullified then that part can continue to exist.” In other words, according to the authors, *shareability implies separability*. A converse confusion appears in (Snoeck & Dedene, 2001), in which the authors claim to present a problematic case of part-whole modeling w.r.t. separability. They use the following example of the relation between a paper and a journal issue: “once a paper has been accepted and published in a journal issue, it has become a part of this issue and is in addition *inseparable from the issue and cannot be shared by other issues*”(p.15). This example shows indeed a case of non-shareability. In fact, one could argue, a case of non-shareability w.r.t. to a given universal (see definition 5.9): although a paper cannot belong to different journals, it could perhaps be part of a *collected papers* series of some given authors.

As a case of non-shareability, this example is perfectly unproblematic, as it can be modeled in a standard way such as in the model of figure 5.32. Anyhow, the important point here is that the paper can be non-shareable without being inseparable. In this case one can perfectly argue that the paper is not *inseparable* from the journal, since it could perfectly well exist prior to the journal existence, and even if the journal issue goes out of existence it could still very well exist as part of the author’s collected paper series⁴¹.

Figure 5-32 Standard UML notation for non-shareable parts

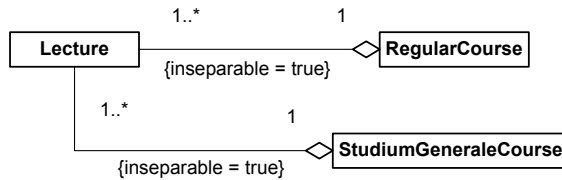


Examples of non-shareability without inseparability abound. To cite just another one: the relation between a Heart to a Human Body is typically conceived as one of non-shareability. Nonetheless, the heart is clearly not

⁴¹ By the terms *Journal Issue* and *Article* here we do not refer to their physical copies (tokens), but to their logical existence.

an inseparable part of the body, since it can survive the extinction of the body. The example actually (fortunately) extends to most human organs, and that is what organ transplants is all about! Moreover, counterexamples of the converse also exist, i.e., there can be shareable parts that are inseparable to all (or some of) the wholes of which they are part. One of such examples is presented in figure 5.33: although a specific lecture of a regular course is an inseparable part of that course, it may be shared with any number of *studium generale* courses. In summary, shareability and separability are orthogonal secondary characteristics.

Figure 5-33 An example of wholes with shareable parts but that are inseparable



Transitivity

Opdahl and colleagues chose to exclude transitivity from the list of primary characteristics of part-whole relations, based on the problematic examples of transitivity paradoxes, such as those discussed in section 5.2.1.

Parthood, as a formal relation, is considered transitive by all theories of parts. In fact, both (Varzi, 2003) and (Simons, 1997) consider transitivity as intrinsic to the very meaning of part. It is true that there are examples of meronymic relations that when put together derive fallacious conclusions. But it is also the case that these meronymic conceptual relations *depart from the formal meaning that mereological binary relations intend to capture* (Johansson, 2004). In a purely mereological sense it is indeed the case that both the door and the handle are parts of house, or that soldiers are parts of battalions. Therefore, a theory of parthood relations for conceptual modeling must explicitly differentiate between the formal relations of parthood defined in formal mereologies and the conceptual relations of parthood that are employed in cognitive tasks. The conceptual relation of parthood should not be interpreted in the same way as its purely formal counterpart.

From a conceptual and computational point of view there are many benefits from reasoning on parthood transitivity. Examples include propagation of properties such as movement, rotation, and (in some cases) creation and destruction, among others. For this reason, on one hand, we agree with Opdahl and colleagues that transitivity is not a “primary characteristic” of *conceptual* part-whole relations, in the sense that it does not hold for all relations of this type. However, on the other hand, we defend that besides assuming non-transitivity for these relations, it is

fundamental to understand why transitivity holds in some cases and not in others, and to determine the *contexts* in which part-whole relations are indeed transitive. Therefore, in this sense, the work presented here can be considered as an extension of the work of Opdahl and colleagues.

5.8 Final Considerations

In this chapter we have concentrated on giving examples of part-whole relations between substantials. However, this is far from implying that this type of relation only holds between individuals pertaining to this ontological category. Conversely, part-whole relationships clearly exist among other types of individuals. Examples of parthood can easily be found among processes and events (perdurants) (buying is part of shopping, singing “happy birthday” is part of birthday party), among conceptual spaces (e.g., the color space is composed of hue, saturation and brightness), moments (e.g., a musical chord is composed of at least three notes; a thought can have proper parts), time and space, etc. Therefore, in this sense, we disagree with the BWW approach (Evermann & Wand, 2001b), which defines parthood only between *things*, i.e., substantial individuals. The categories of perdurants, moments and conceptual spaces are defined in chapter 6.

The focus on substantials here is motivated by the main objectives of this thesis, which emphasizes on providing ontological foundations for structural conceptual modes. Nonetheless, we strongly believe that most of the results developed here can be carried to part-whole relations holding among individuals of different categories. For example, the basic axioms of MM also hold for parts of processes, which also have mandatory and contingent parts, and so forth. For a study of the part-whole relation among processes, we refer to (Simons, 1987).

One of the main objectives for the development of the typology of meronymic relations proposed by Winston, Chaffin and Herrmann (WCH) is to address the problem of fallacious transitive cases. As demonstrated in section 5.6, the typology of parthood relation that can be extracted from a revision of WCH’s original account can be exploited to provide more direct guidelines for helping the user in the definition of these contexts. In chapter 7, we resume the discussion on this topic by proposing a number of visual patterns that can be used to identify transitivity context in parthood relations among complexes, which are the most typical sort of parthood relations to be found in conceptual modeling in computer science.

We believe that the problem of transitivity bears some strong connection to another (very difficult) philosophical question, namely, that of why are some parts essential and other parts accidental to a thing. As

discussed in (Simons, 1987; Wiggins, 2001), the parts of a thing behave much like any other property of the thing. For this reason, parts are also determined by the nature of the particular or, to put in Aristotelian terms, by “what the particular is”? Since, in conformance with Wiggins, we consider that the *essential properties* of a thing are those implied by the (unique) *substance sortal* that it instantiates, the same applies to its essential parts.

This makes clear that any deeper discussion on the reasons for the essentiality of some parts requires a discussion on the essentiality of properties. In our opinion, such a discussion cannot be carried out avoiding the philosophical discussion on the very meaning of Kinds, which in turn, is related to the ancient issues of essentialism and the problem of universals (Armstrong, 1989). Some authors (most notably Boyd, 1991), deny the existence of essences in traditional terms (i.e., as a list of essential characteristics) in favour of what can be termed *nomological essentialism*. Boyd proposes a view of essence in terms of what he names *causal homeostasis*. For him, the essence of a thing is a network of *causal dispositions* relating the properties of the thing. He argues that: “kinds, properties, relations, etc. are natural if they reflect important features of the causal structure of the world... the clustering of properties is not merely chance coincidences...the presence of some properties tends to favor the presence of others where there are common underlying properties that tend to maintain the presence of the clusters.” (cited by Keil, 1992).

This notion of essence as a cluster of causally related properties is akin to the notion of *organization* proposed by Maturana and Varela (Maturana & Varela, 1987; Varela, 1979) in their theory of Autopoiesis (as noticed by Simons, 1998). This view implies that an individual can change all its manifest properties (e.g., perceptual characteristics) and keep its identity as long as its organization is maintained. Actually, Boyd’s argument is even stronger than that: some properties of the thing have to vary for the *homeostasis* to be maintained. This position is also shared by Bunge who states that the laws of a thing are essential to it (by definition of a law, it must hold necessarily), and the necessary properties of a thing are those which are lawfully related. In summary, both laws and lawfully related properties are essential. Since (i) the laws of an object are determined by its Natural Kind (Bunge, 1977, chapter 3), and (ii) a Natural Kind in Bunge is equivalent to what is termed here *substance sortal* (see discussion in section 4.5.2), we can conclude that indeed the necessary properties of a thing are determined by the (unique) substance sortal it instantiates.

In discussing individual adaptation, Bunge and more recently (Rowe & Leaney, 1997) and (Rowe & Leaney & Lowe, 1998) argue about change of properties in terms of Bunge’s concept of the *nomological state space* of a thing. The idea is that a thing can only assume states that are contained in

its nomological state space, i.e., *lawful states*. Suppose that a property p of a thing x changes, placing it in an unlawful state s . This change must trigger changes in other property(ies), say p' , so that another lawful state s' can be reached. If no such compensatory change is possible, then object x will remain in a state that falls outside its nomological state space and, according to Bunge, “it will change its name”. In our terminology, this means that x will change its identity and therefore will cease to exist as such. As a consequence, we can state that the essential parts of a thing are those that are directly related to the thing's identity and, hence, those whose loss puts the thing in an unrecoverable unlawful state, i.e., in a state where no compensatory change exists to bring it back to its nomological state space.

This conception of essence finds support in many authors in cognitive sciences (Keil, 1992; Milikan, 1998) and it has been empirically supported by works such as (Schunn & Vera, 1995). In particular, Schunn & Vera provide some empirical evidence that human cognition employs *causal domain theories* (and not typicality or frequency of properties) as its most important principle of application, and this is the case for both endurants and perdurants, natural entities and artifacts, and for entities of familiar and unfamiliar kinds.

The cognitive psychologist Frank Keil (Keil, 1992) subscribes to the stronger idea that *homostatic property clusters* exist not only for natural but also for nominal kinds (Schwartz, 1980). He argues that it is unlikely that the elements in a nominal essence are fully arbitrary and defends that, in contrast, they are systematically connected to a set of real, or at least supposed, causal relations – not those in biology or physics but those governing human interactions. For instance, chairs have a number of properties, which are used to identify them. Although there may not be internal causal homeostatic mechanisms of chairs that lead them to have these properties, there may well be external mechanisms having to do with the form and function of the human body and with typical social and cultural activities of humans. For example, certain dimensions of chairs as well as the weight range they are supposed to support are determined by properties of the human anatomy. Some of the features of nominal kinds can be arbitrary. For instance, take the rules of traffic: the choice of driving on the right or left of the road seems a random choice. However, having a rule that decides which side to drive it is not, since there are good causal justifications related to the consequence of collisions to humans on certain velocities and the difficulties in responding quickly enough with such a rule at certain velocity. In summary, for Keil, although the definition of nominal kinds (including artifacts) is heavily dependent of human intentions, the latter are not in themselves arbitrary. According to him, perhaps a clearer distinction between natural and nominal kind is that for the former the causal homeostatic mechanisms are closely related to various domains of

science (e.g. biology, chemistry, physics), whereas for the latter social and psychological domains of causality are involved.

According to (van Leeuwen, 1991), the principle of identity and, in particular, the principle of persistence that an individual obeys is implied by its principle of application. This is compatible with his view that endurants are *persisting integral wholes*, i.e., if the principle of application is determined by the presence of a *causal network of properties* (essence) then a continuant is an object that persists in time maintaining that network, or as system theorists would say, maintaining its *organization*. Moreover, in parity with Bunge, the individual *is the same* as long as this network is maintained. Deciding what is the *integral whole* that should persist is therefore the very task of the principle of application. This supports the thesis that the *principle of unity* of an individual is also strongly related to its principles of application and identity. In order to individuate something we must decide what its parts are, i.e., we must see it as an *integral whole*. Thus counting presupposes individuation, individuation presupposes unity and unity presupposes application. Therefore, it is no coincidence that the universals that carry principles of identity, persistence, individuation and counting are also exactly those that also carry a principle of unity, namely sortal universals.

Finally, if the principle of unity for an individual *x* is related to its principle of application (and so is its *unifying condition*), to determine “what *x* is” helps to determine what its parts are. In particular, if a principle of application is a causal network of properties then to determine *x*’s parts is to determine the particulars that contribute to this network. Ergo, objects that are unified under the same unity condition are those that contribute to the causal network of properties that supply this condition. This argument seems to support the thesis defended in this chapter that transitivity holds within wholes that are unified by (refinements of) the same unity condition. In other words, material part-whole relations can be seen as embedding the implicit condition Φ (see discussion in section 6.2.7) that their relations contribute to the same cluster of properties (nomological essence).

A number of results in cognitive psychology provide empirical evidence for some of the points aforementioned. For example, (Tversky, 1989) shows that taxonomies and partonomies are strongly interrelated, and that children are more likely to group taxonomically when instances share parts than when instances do not share parts. This clearly suggests the relation between a criterion of unity and one of application. Moreover, as discussed in chapter 4, the establishment of kind categories (substance sortals) in the language of thought is essential for the learning of proper names, common nouns and for individuation and categorization tasks in general. According to (Mcnamara, 1986), the first notion of kind which young infants construct is in association with the so-called *basic-level sortals* (Rosch et al., 1976). In taxonomy, basic-level sortals have a privileged status in many

cognitive tasks, and it is at this level that subjects are first able to substantially produce attributes for the given categories. (Tversky & Hemenway, 1984) shows that most of these attributes are parts, and propose that parts “may underlie the informativeness of the basic level and account for the convergence of many cognitive operations at that level as well”. Finally, (Hall, 1998) conducts empirical studies with regard to the perceived conceptual relation between the identity of an object and the change of its parts. As his results show, “knowledge about *specific kinds of objects* and their canonical transformations exerts an increasingly powerful effect, over the course of development, upon people’s tendency to rely on continuity as a criterion for attributing persistence to objects that undergo change.”

Finally, we summarize this discussion from a more pragmatic conceptual modeling point of view. Recently, many authors have proposed that an interesting way to conceptualize structure/organization of entities is to view it as a set of interrelated roles (Dignum, 2003; Odell & Parunak & Fleischer, 2003; Odell & Nodine & Levy, 2005). This possibility is also accounted for in (Odell & Bock, 1998), in which integral wholes are claimed to be “*aggregates of qua types*”. Although we mostly agree with this conception, there is still one point that should be rephrased. As discussed in chapter 4, roles are anti-rigid universals, and stating that wholes are unified by objects that play roles in its inner structure implies that all objects are only contingently part of this structure. We advocate that this is the case for some objects, namely mandatory parts⁴². However, for some objects this relation is not contingent, in which case one could say that they “play the roles” of themselves. These are the inseparable and/or essential parts. In summary, an object *x* is part of another object *y* if and only if it “plays a role” defined in the organization (context) of *x*.

We formally define the notion *qua individuals* in chapter 7. Moreover, in that chapter, we resume the discussion on the connection between *qua individuals* and part-whole relations.

⁴²Theoretically they could also be merely contingent parts but this would demand the specification of zero minimum cardinalities in the (part-whole) relations between the aggregate and the role these parts play. This in turn would entail the creation of invalid models according to the postulates of chapter 4, since roles are relationally dependent entities.